
Automotive SoC Motor Driver

1.0 GDU (GATE DRIVER UNIT)

1.1 Features

- 3-phase gate driver for external MOSFET
- Integrated gate slew rate control
- Dedicated charge pump for low-battery operation
- VCP charge pump for full range duty cycle operation
- On-chip 5V LDO for MCU power supply
- Battery voltage compliant external sensor terminal
- Q&A watchdog with dedicated interface to MCU
- Dedicated analog voltage acquisition
- Dedicated full differential amplifier for low-side shunt current measurement
- Single ended current sensing amplifier output (Only available option A and C)
- battery compliant Fault output (Only available option B and D)
- Battery voltage compliant ignition terminal
- LIN transceiver LIN2.x/ISO17987/J2602 compatible
- Dedicated BEMF / Phase comparator
- On-chip dedicated NVM (EEPROM) for user storage
- External enable input to enable the gate driver

1.2 Introduction

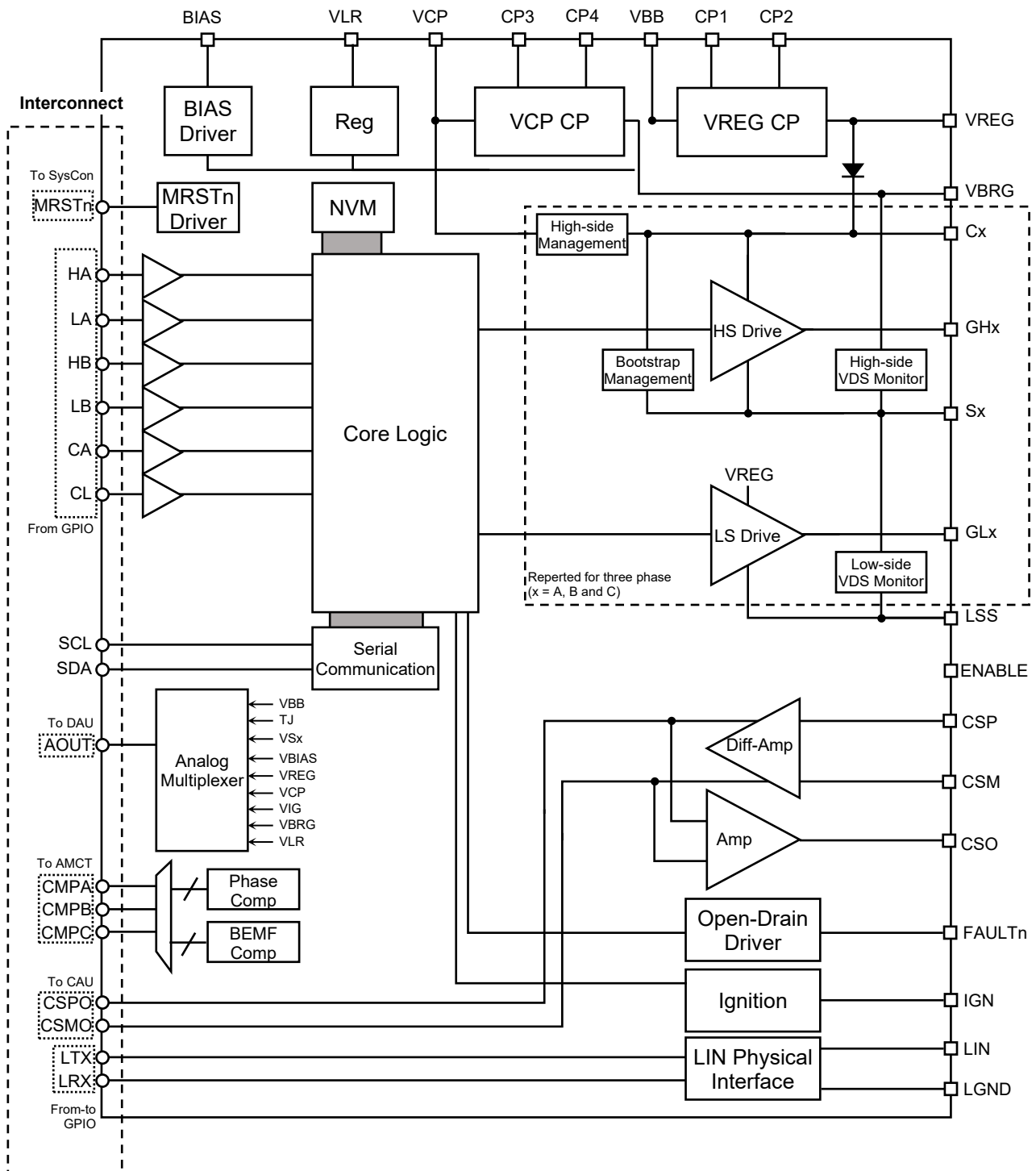
The gate driver is an N-channel power MOSFET driver capable of controlling MOSFETs connected in a 3-phase bridge arrangement and is specifically designed for automotive power applications with high-power inductive loads, such as BLDC motors.

A unique charge pump regulator provides the supply for the MOSFET gate drive for battery voltages down to 7V and allows the A89201 to operate with a reduced gate drive voltage down to 5.5V. Gate drive voltage and strength are programmable to help reduce EMC issues. A bootstrap capacitor is used to provide the above-battery supply voltage required for N-channel MOSFETs. Bootstrap charge is maintained at high PWM duty cycles by an additional charge pump providing 0-100% PWM with no duty cycle restriction.

The power supply unit provides and manages all internal supplies from a single 5.5 to 50V supply. The MCU section can also operate with an independent single 5V supply.

Integrated programmable diagnostics provide indication of multiple internal faults, system faults, and power bridge faults, and can be configured to protect the power MOSFETs under most short circuit conditions.

2.0 Functional Description



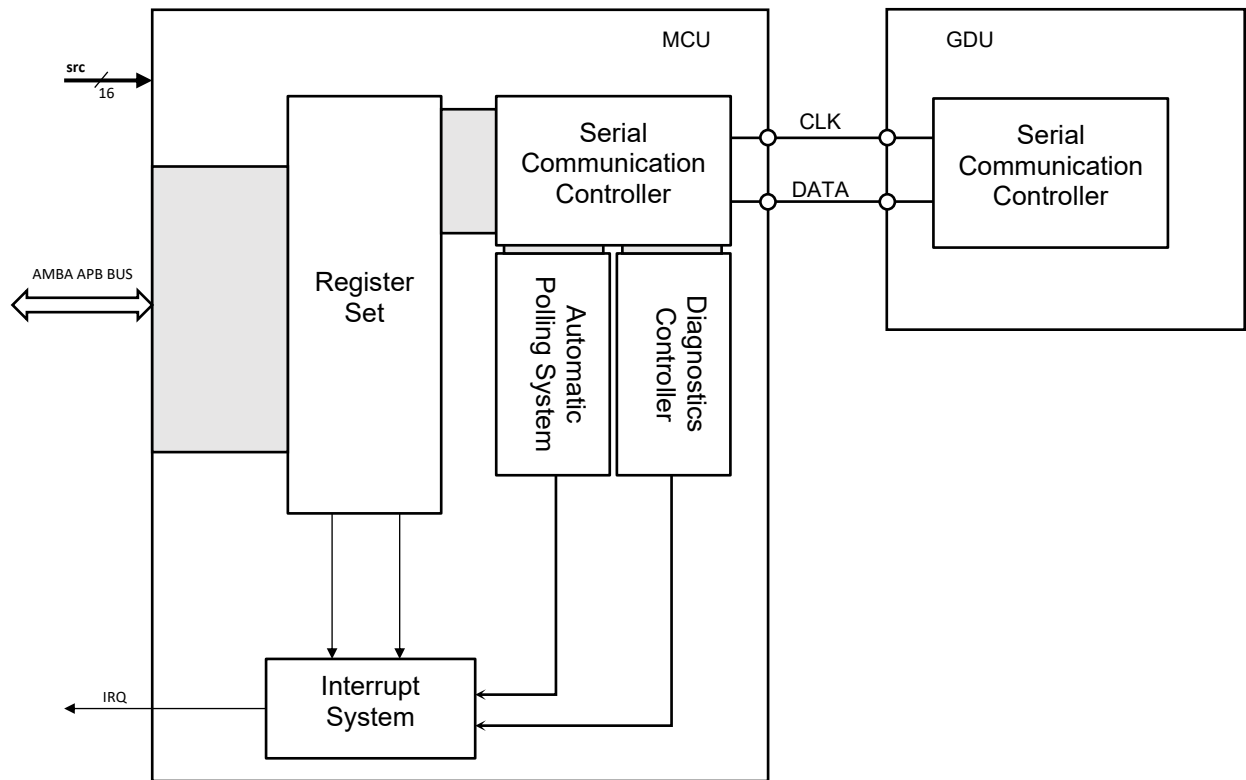


Figure 1-1: Block diagram of MCU and GDU communication

2.1 Register Set

GDU has a control register that can configure

This section describes the dedicated MCU peripheral to control the serial communication.

2.1.1 Register Map

The register map of the GDU serial communication consists of 9 16-bit registers. Table 1 summarizes the register addresses, access types and reset values. The starting address of the GDU serial communication controller module is 0x40001000.

Base Address: 0x40001000

Table 1: Register map of the GPIO module

Register Name	Access Type	Address Offset	Reset Value
addr		0x00	0x0000
tx		0x04	0x0000
rx		0x08	0x0000
ctrl		0x0C	0x0000
inten		0x10	0x0000
intstat		0x14	0x0000
tcf		0x18	0x0000
iginten		0x1C	0x0000
igntstat		0x20	0x0000

2.1.2 Register Definition

ADDR Register

Reset value = 0x0000

Address Offset: 0x00

When the ADDR register is written, an transfer will be generated on the serial communication interface. If the serial communication is busy with the auto-read of the TCF then the transfer is deferred until the TCF transfer is complete.

When the ADDR register is written the transfer becomes pending and remains so until the transfer is completed, If a second ADDR write occurs whilst the first is still pending it is ignored and the OVF status bit is set, and if enabled an OVF interrupt is also raised.

Note that the ADDR register refers to the GDU register address as in the GDU configuration register specification.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	-	-	-	-
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
addr							rnw
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
rnw	0	Read/Write selector 0: Write Operation 1: Read Operation
addr	7:1	GDU Address register

Tx Register

Reset value = 0x0000

Address Offset: 0x04

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
data[15:8]							
RW	RW	RW	RW	RW	RW	RW	RW

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
data[7:0]							
RW	RW	RW	RW	RW	RW	RW	RW

Legend:

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
data	15:0	16bit transmit data

Rx Register

Reset value = 0x0000

Address Offset: 0x08

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
data[15:8]							
R	R	R	R	R	R	R	R

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
data[7:0]							
R	R	R	R	R	R	R	R

Legend:

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
data	15:0	16bit received data

CTRL Register

Reset value = 0x0000

Address Offset: 0x0C

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
			freq			mode	
			RW	RW	RW	RW	RW

Legend:

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
mode	1:0	Mode selection 00: Disabled 01: User Operation enabled 10: Disabled 11: Automatic polling and user operation enabled
freq	4:2	Division of clock to generate serial communication clock and polling period

	SCL	Polling Period
000	10MHz	25us
001	5MHz	50us
010	2MHz	125us
011	1MHz	250us
Reserved	N/A	N/A
1XX	10MHz	25us

Note:

- 1- When CPU clock, *mclk* is changed the serial communication clock will be changed accordingly. Please refer SCU section of MCUCCLK register.
- 2- Do not use Reserved value.

INTEN Register

Reset value = 0x0000

Address Offset: 0x10

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	-	-	-	-
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-	-	ff	ovf	crc	ack	rx	tx
U	U	RW	RW	RW	RW	RW	RW

Legend:

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
tx	0	Sending complete interrupt enable 0: Disabled 1: Enabled
rx	1	Receiving complete interrupt enable 0: Disabled 1: Enabled
ack	2	Serial Communication Acknowledge error interrupt enable 0: Disabled 1: Enabled
crc	3	CRC error interrupt enable 0: Disabled 1: Enabled
ovf	4	Pending buffer overflow interrupt enable 0: Disabled 1: Enabled
ff	5	GDU fault flag interrupt enable 0: Disabled 1: Enabled

INTSTAT Register

Reset value = 0x0000

Address Offset: 0x014

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
polling	busy	ff	ovf	crc	ack	rx	tx
R	R	R1C	R1C	R1C	R1C	R1C	R1C

Legend:

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
tx	0	Sending complete interrupt 0: Not active 1: Active
rx	1	Receiving complete interrupt 0: Not active 1: Active
ack	2	Serial Communication Acknowledge error interrupt 0: Not active 1: Active
crc	3	CRC error interrupt 0: Not active 1: Active
ovf	4	Pending buffer overflow interrupt 0: Not active 1: Active
ff	5	GDU fault flag interrupt 0: Not active 1: Active
busy	6	Busy indicates a user transmit transfer is pending or in progress 0: No user transmit transfer 1: Busy <u>Note: This bit is not latched. When the transmission complete, it will show as 0.</u>
polling	7	Polling indicates an automatic polling transfer is pending or in progress 0: No automatic polling transfer 1: Busy <u>Note: This bit is not latched. When the transmission complete, it will show as 0.</u>

TCF Register

TCF (Timing critical Register) is updated when GDU address 127 is read by either automatic polling or manual reading. FF and IG Register are the trigger interrupton for CPU.

Reset value = 0x0000

Address Offset: 0x018

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
FF	POR	SE	EE	DSO	IG	OT	TW
R	R	R	R	R	R	R	R

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
BU	WD	ETO	SBOU	VCPU	VBU	VRU	VLRU
R	R	R	R	R	R	R	R

Legend:

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
FF	15	General fault flag FF = POR SE EE DSO OT TW BU WD ETO SBOU VCPU VBU VRU VLRU FF bit is set when the others fault is flagged except IG. 0: Not active 1: Active Note: FF active can generate interrupt to CPU when INTEN.ff is enabled.
POR	14	Power on Reset 0: Inactive 1: Active Please refer the datasheet for further detail of POR.
SE	13	Serial Communication Error 0: Not active 1: Fault detected This bit is flagged when each of following bits active: ESTO, ENV, ESZ, EAC, ECRC
EE	12	NVM Error 0: Not active 1: Fault detected
DSO	11	VDS over voltage 0: Not active 1: Fault detected It is flagged when each of following bits active: AHO, ALO, BHO, BLO, CHO, CLO,
IG	10	Ignition Input state 0: Low (Less than IG low threshold) 1: High (Higher than IG high threshold) Note: IG High can generate interrupt to CPU when it is enabled on IGINTEN.ig.
OT	9	Over Temperature 0: Not active 1: Fault detected
TW	8	Temperature Warning 0: Not active 1: Fault detected
BU	7	Bootstrap Undervoltage 0: Not active 1: Fault detected It is flagged when each of following bits active: VC, VB, VA
WD	6	Q&A watchdog fault 0: Not active 1: Fault detected

ETO	5	ENABLE watchdog timeout 0: Not active 1: Fault detected
SBOU	4	Sensor BIAS terminal undervoltage 0: Not active 1: Fault detected
VCPU	3	VCP Undervoltage 0: Not active 1: Fault detected
VBU	2	VBB Undervoltage 0: Not active 1: Fault detected
VRU	1	VREG Undervoltage 0: Not active 1: Fault detected
VLRU	0	VLR Undervoltage 0: Not active 1: Fault detected

IGINTEN Register

Reset value = 0x0000

Address Offset: 0x01C

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	-	-	-	-
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-	-	-	-	-	-	-	ig
U	U	U	U	U	U	U	R/W

Legend:

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
ig	0	Ignition Interrupt Enable 0: Disabled 1: Enabled

IGINTSTAT Register

Reset value = 0x0000

Address Offset: 0x020

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
							ig
							R1C

Legend:

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

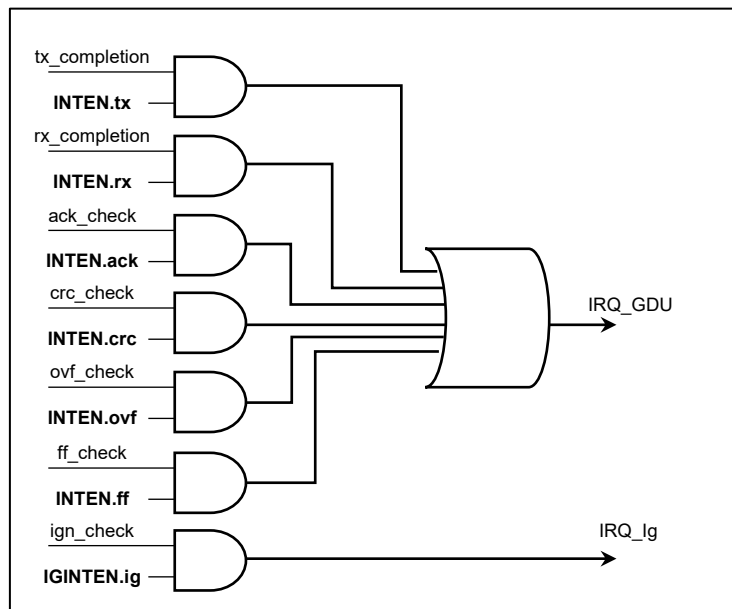
Res: Reserved bit

Field	Bits	Description
ig	0	Ignition interrupt 0: No interrupt request 1: Interrupt is enabled Note: This flags when TCF.data[10] is set.

2.2 Interrupt generation and management

2.2.1 Interrupt control system

Interrupt generation logic is as follows:



GDU interrupt will be only enabled when the bit in INTEN register is enabled.

Ignition interrupt is enabled only when IGNITEN.ig is set to high. When the IRQ routine is called, the status bit shall be cleared by writing 1 to INTSTAT or IGINTSTAT register.

2.2.2 Acknowledge Check

When the data is transmitted to the GDU from the MCU, the acknowledge bit is added to confirm the serial communication transaction. When the acknowledge is failed, the communication will be cancelled, and the data transmitted will be discarded.

When the acknowledgement error is detected, it will be flagged to *INTSTAT.ack*. If the IRQ is enabled though INTEN.ack, then GDU_IRQ will be called.

2.2.3 CRC Check

In each serial communication frame, CRC frame is added. This improve the serial communication robustness.

If CRC check is failed, the received data is ignored and the RX register is not updated. When it is detected, *INTSTAT.crc* will be flagged.

2.3 GDU Register

The GDU registers from address 0 to 40 and address 123 will be stored into NVM when programmed though WNV. The other register will not be programmed. The GDU configuration register access must be performed by GDU serial communication control interface.

Address	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	VCPM			VDQ	VQT						VT					
	0	0	0	0	0	1	0	0	0	0	0	1	1	0	0	0
1	CPM							OSO	SC	SAO				SAG		
	0	0	0	0	0	0	0	0	0	1	0	1	0	0	1	0
2	AMXB	AMX								SBS	SBC			SBM		
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3														DAG		
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
4					TRS				IR1				IR2			
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5					TFS				IF1				IF2			
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
6	TTEN	DWL			OPM	LBB	LEN							BM		
	0	0			0	0	0	0	0	0	0	0	0	0	0	0
7	FSE		FDSO		FTW	FOT		FWD	FEN		FVBU		FVCP U	FBU	FVRU	
	1		1		1	1		1	1		1		1	1	1	
8	OT	TW	VCPU	VBU	VRU		SBOV	ETO	WD	BSU	CHO	CLO	BHO	BLO	AHO	ALO
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
9-40	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
120	TTO															
	0	0	0	0	-	-	-	-	-	-	-	-	-	-	-	-
121							RNV	WNV								
							0	0	0							
122	FLT			GTS						ENBB	CH	CL	BH	BL	AH	AL
	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
123			WC				WL				WD					
			0	0	0	0	0	1	0	1	0	0	1	0	1	0
124	QA															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
125	EEB	VLR							SC	GDO	CHM	CLM	BHM	BLM	AHM	ALM
126				ENV	ESZ	EAC	ECRC	VC	VB	VA	CHO	CLO	BHO	BLO	AHO	ALO
127	FF	POR	SE	EE	DSO	IG	OT	TW	BU	WD	ETO	SBOU	VCPU	VBU	VRU	VLU

Driver Register

Reset value = 0x0418

GDU Address: 0x00

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
VCPM		-	VDQ	VQT[5:2]			
RW	RW	RW	RW	RW	RW	RW	RW

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
VQT[1:0]		VT					
RW	RW	RW	RW	RW	RW	RW	RW

Legend:

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
VT	5:0	VDS overvoltage threshold $V_{DS_TH} = n * 0.05 [V]$ Default: $V_{DS_TH} = 1.2V$ (n = 011000)
VQT	11:6	VDS Quorification time $t_{VDQ} = n * 0.1 [\mu s]$ Default: $t_{VDQ} = 1.6[\mu s]$ (n = 010000)
VDQ	12	VDS overvoltage fault quarifier mode 0: Debounce 1: Blank
VCPM	15:14	VCP chargepump Mode 00: Auto 500mA/200μA 01: 200uA 10: 5mA 11: Disabled VCP charge pump

Sense Select Register

Reset value = 0x0052

GDU Address: 0x01

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
CMP							OSO
RW	RW	RW	RW	RW	RW	RW	RW

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
SC	SAO				SAG		
RW	RW	RW	RW	RW	RW	RW	RW

Legend:

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
CMP	15:13	Select Comparator Output 000: BEMF comp (Default) 001: Phase comp 010: Drive Low 011: Drive High 1xx: Drive low
OSO	8	CSO output control 0: Normal Mode 1: Offset output mode Note: In offset mode, differential input is ignored, and output only offset.
SC	7	Calibrate Sensing amplifier 0: No action 1: No action 1->0: No action 0->1: Initiate Calibration
SAO	6:3	Sense amplifier offset 0000: 0V 0001: 0V 0010: 0.1V 0011: 0.1V 0100: 0.2V 0101: 0.3V 0110: 0.4V 0111: 0.5V 1000: 0.75V 1001: 1.0V 1010: 1.2V (Default) 1011: 1.5V 1100: 1.65V 1101: 2.0V 1110: 2.25V 1111: 2.4V
SAG	2:0	Sense amplifier gain 000: 10 001: 16 010: 20 (Default) 011: 26 100: 30 101: 36 110: 40 111: 50

BIAS terminal and AMUX control Register

Reset value = 0x0

GDU Address: 0x02

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
AMXB	AMX						
RW	RW	RW	RW	RW	RW	RW	RW

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
	SBS	SBC			SBM		
RW	RW	RW	RW	RW	RW	RW	RW

Legend:

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
AMXB	15	Analog Mux Buffer Output 0: No buffer (Default) 1: Buffer used
AMX	14:11	Analog Multiplexer Output select 0000: VBB (Default) 1000: Tj (GDU die temp) 0001: VBRG 1001: IG 0010: SA 1010: OOS 0011: SB 1011: BIASRB 0100: SC 1100: Unused 0101: VREG 1101: GND 0110: VCP 1110: GND 0111: VLR 1111: GND
SBS	6	Sensor BIAS terminal sleep 0: Functional (Default) 1: BIAS terminal in sleep mode
SBC	5:3	Sensor Bias Current 000: Zero (Leakage only) (Default) 100: 250 µA 001: 25 µA 101: 500 µA 010: 50 µA 110: 500 µA 011: 100 µA 111: 2.5mA
SBM	2:0	Sensor BIAS Mode 0XX: Disabled (High Impedance) (Default) 100: Current Mode 101: Current Mode 110: Voltage Mode 111: Voltage Mode

Differential Sensing Amplifier Control Register

Reset value = 0x0002

GDU Address: 0x03

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
RW	RW	RW	RW	RW	RW	RW	RW

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
						DAG	
RW	RW	RW	RW	RW	RW	RW	RW

Legend:

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
DAG	2:0	Differential Amplifier Gain 000: 5 001: 8 010: 10 (Default) 011: 13 100: 15 101: 18 110: 20 111: 25

Slew Rate Control for Rising Register

Reset value = 0x0000

GDU Address: 0x04

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
				TRS			
RW	RW	RW	RW	RW	RW	RW	RW

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
IR1				IR2			
RW	RW	RW	RW	RW	RW	RW	RW

Legend:

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
TRS	11:8	Slew control turn-on time for rising $t_{ris} = n * 50 \text{ [ns]}$ Default: $t_{ris} = 0 \text{ [ns]}$ (Disabled)
IR1	7:4	Turn-on Current 1 for rising $I_{ris_1} = n * 16 \text{ [mA]}$ $I_{ris_1} = 0 \text{ [mA]}$ (Default: Disabled)
IR2	3:0	Turn-on Current 2 for rising $I_{ris_2} = n * 16 \text{ [mA]}$ $I_{ris_2} = 0 \text{ [mA]}$ (Default: Disabled)

Slew Rate Control for Falling Register

Reset value = 0x0000

GDU Address: 0x05

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
RW	RW	RW	RW	RW	RW	RW	RW

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
RW	RW	RW	RW	RW	RW	RW	RW

Legend:

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
TFS	11:8	Slew control turn-on time for falling $t_{fal} = n * 50$ [ns] Default: $t_{fal} = 0$ [ns] (Disabled)
IF1	7:4	Turn-off Current 1 for falling $I_{fal_1} = n * 16$ [mA] $I_{fal_1} = 0$ [mA] (Default: Disabled)
IF2	3:0	Turn-off Current 2 for falling $I_{fal_2} = n * 16$ [mA] $I_{fal_2} = 0$ [mA] (Default: Disabled)

GDU Config Register

Reset value = 0x0000

GDU Address: 0x06

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
TTEN	DWL			OPM	LBB	LEN	
RW	RW	RW	RW	RW	RW	RW	RW

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
RW	RW	RW	RW	RW	RW	RW	RW

Legend:

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
TTEN	15	Tiny Temp Enable 0: Disabled 1: Enabled
DWL	14	Disable Wake on LIN terminal 0: Active 1: Disabled
OPM	11	Operating Mode Select 0: LIN Mode 1: PWM Mode
LBB	10	LIN Baud Rate 0: 10kHz 1: 20kHz
LEN	9	LIN Enable 0: Standby (Recessive) 1: Active

SoF Control Register

Reset value = 0xADAE

GDU Address: 0x07

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
FSE		FDSO		FTW	FOT		FWD
RW	RW	RW	RW	RW	RW	RW	RW

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
FEN		FVBU		FVCPU	FBU	FVRU	
RW	RW	RW	RW	RW	RW	RW	RW

Legend:

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
FSE	15	Serial Communication Error
FDSO	13	VDS Overvoltage Error
FTW	11	Temperature Warning
FOT	10	Over Temperature
FWD	8	Q&A Watchdog
FEN	7	Enable Time-out
FVBU	5	VBB Undervoltage
FVCPU	3	VCP Undervoltage
FBU	2	Bootstrap Undervoltage
FVRU	1	VREG Undervoltage

Fault Mask Control Register

Reset value = 0x0000

GDU Address: 0x08

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
OT	TW	VCPU	VBV	VRU		SBOU	ETO
RW	RW	RW	RW	RW	RW	RW	RW

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
WD	BSU	CHO	CLO	BHO	BLO	AHO	ALO
RW	RW	RW	RW	RW	RW	RW	RW

Legend:

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
OT	15	Over temperature
TW	14	Temperature Warning
VCPU	13	VCP Undervoltage
VBV	12	VBB undervoltage
VRU	11	VREG undervoltage
SBOU	9	Sensor BIAS undervoltage
ETO	8	ENABLE time-out
WD	7	Q&A Watchdog error
BSU	6	Bootstrap Undervoltage
CHO	5	Phase C high-side VDS overvoltage
CLO	4	Phase C Low-side VDS overvoltage
BHO	3	Phase B high-side VDS overvoltage
BLO	2	Phase B Low-side VDS overvoltage
AHO	1	Phase A high-side VDS overvoltage
ALO	0	Phase A Low-side VDS overvoltage

User Data area Register

Reset value = 0x0000

GDU Address: 0x09 – 0x28

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
Data[15:8]							
RW	RW	RW	RW	RW	RW	RW	RW

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Data[7:0]							
RW	RW	RW	RW	RW	RW	RW	RW

Legend:

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
Data	15:0	User Data Note: This area is provided to store user data. No effect for the GDU function.

NVM Control Register

Reset value = 0x0000

GDU Address: 0x79

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
						RNV	WNV[1]
RW	RW	RW	RW	RW	RW	RW	RW

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
WNV[0]							
RW	RW	RW	RW	RW	RW	RW	RW

Legend:

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
RNV	9	Read Non-volatile memory 0: No action 1: No action 0->1: Read NVM and update registers 1->0: No action
WNV	8:7	Write Non-volatile Memory When WNV[1:0] is changed from 01 to 10 the present contents of registers from 0 – 40 will be written to NVM. When the NVM save has completed successfully WNV[1:0] will be set to 01 and can be read to verify completion of the write. If WNV[1:0] is reset to 00, the save has not completed successfully.

Gate Drive Control Register

Reset value = 0x8000

GDU Address: 0x7A

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
FLT			GTS				
RW	RW	RW	RW	RW	RW	RW	RW

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
	ENBR	CH	CL	BH	BL	AH	AL
RW	RW	RW	RW	RW	RW	RW	RW

Legend:

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
FLT	15	FAULTn terminal driver 0: Open-drain Transistor turn-off 1: Open-drain Transistor turn-on
GTS	12	Go to sleep 0: No action 1: No action 1->0: No action 0->1 Enter sleep if enabled
ENBR	6	Enable Bridge 0: Bridge disabled (All gate is low) 1: Bridge enabled
CH	5	Command for C-phase high-side 0: Off 1: Turn-on the gate
CL	4	Command for C-phase low-side 0: Off 1: Turn-on the gate
BH	3	Command for B-phase high-side 0: Off 1: Turn-on the gate
BL	2	Command for B-phase low-side 0: Off 1: Turn-on the gate
AH	1	Command for A-phase high-side 0: Off 1: Turn-on the gate
AL	0	Command for A-phase low-side 0: Off 1: Turn-on the gate

Q&A Watchdog Configuration Register

Reset value = 0x14A

GDU Address: 0x7B

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
		WC				WI[4:3]	
RW	RW	RW	RW	RW	RW	RW	RW

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
WI[2:0]			WD				
RW	RW	RW	RW	RW	RW	RW	RW

Legend:

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
WC	13:10	Q&A Watchdog fail cycle count before sleep $N_{fail_cy} = n$ [Cycle] Default: $N_{fail_cy} = 0$ Note: When $n = 0$, sleep feature is disabled and remain the GDU is powered up. Only bridge will be disabled.
WI	9:5	Q&A Watchdog Initial Wait time $T_{WI} = (1+n)$ [ms] Default: $T_{WD_wait} = 11$ [ms]
WD	4:0	Q&A Watchdog time out $T_{timeout} = (4n + 1)$ [ms] Default: $T_{timeout} = 41$ [ms]

Q&A Watchdog Control Register

Reset value = 0x0000

GDU Address: 0x7C

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
QA[15:8]							
RW	RW	RW	RW	RW	RW	RW	RW

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
QA[7:0]							
RW	RW	RW	RW	RW	RW	RW	RW

Legend:

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description																																		
QA	15:0	Q&A Control Register When QA register is read-out, QA[3:0] is the address for the Q&A watchdog. When QA[15:0] is written, 16bit answer code is required to clear the watchdog timer. <table><tr><th>Address[3:0]</th><th>Answer code[15:0]</th></tr><tr><td>0x0</td><td>0xCDF2</td></tr><tr><td>0x1</td><td>0xF738</td></tr><tr><td>0x2</td><td>0xA847</td></tr><tr><td>0x3</td><td>0xF8F8</td></tr><tr><td>0x4</td><td>0x1BD5</td></tr><tr><td>0x5</td><td>0x2414</td></tr><tr><td>0x6</td><td>0xE524</td></tr><tr><td>0x7</td><td>0xFBC7</td></tr><tr><td>0x8</td><td>0x47E3</td></tr><tr><td>0x9</td><td>0x6055</td></tr><tr><td>0xA</td><td>0x271B</td></tr><tr><td>0xB</td><td>0x4C21</td></tr><tr><td>0xC</td><td>0x3033</td></tr><tr><td>0xD</td><td>0x19EB</td></tr><tr><td>0xE</td><td>0x7D10</td></tr><tr><td>0xF</td><td>0x971B</td></tr></table>	Address[3:0]	Answer code[15:0]	0x0	0xCDF2	0x1	0xF738	0x2	0xA847	0x3	0xF8F8	0x4	0x1BD5	0x5	0x2414	0x6	0xE524	0x7	0xFBC7	0x8	0x47E3	0x9	0x6055	0xA	0x271B	0xB	0x4C21	0xC	0x3033	0xD	0x19EB	0xE	0x7D10	0xF	0x971B
Address[3:0]	Answer code[15:0]																																			
0x0	0xCDF2																																			
0x1	0xF738																																			
0x2	0xA847																																			
0x3	0xF8F8																																			
0x4	0x1BD5																																			
0x5	0x2414																																			
0x6	0xE524																																			
0x7	0xFBC7																																			
0x8	0x47E3																																			
0x9	0x6055																																			
0xA	0x271B																																			
0xB	0x4C21																																			
0xC	0x3033																																			
0xD	0x19EB																																			
0xE	0x7D10																																			
0xF	0x971B																																			

Diagnostic 0 Register

Reset value = 0xC800

GDU Address: 0x7D

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
EEB							
R	R	R	R	R	R	R	R

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
SC	GDO	CHM	CLM	BHM	BLM	AHM	ALM
R	R	R	R	R	R	R	R

Legend:

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
EEB	15	NVM Busy Indication 0: Free 1: Busy
SC	7	System Clock 0: Running 1: Stop
GDO	6	All gate drive off 0: Not off 1: Off
CHM	5	Phase-C high-side monitor 0: $V(\text{VBRG}, \text{SC}) < (V_{\text{DS_TH}})$ 1: $V(\text{VBRG}, \text{SC}) > (V_{\text{DS_TH}})$
CLM	4	Phase-C Low-side monitor 0: $V(\text{SC}, \text{LSS}) < (V_{\text{DS_TH}})$ 1: $V(\text{SC}, \text{LSS}) > (V_{\text{DS_TH}})$
BHM	3	Phase-B high-side monitor 0: $V(\text{VBRG}, \text{SB}) < (V_{\text{DS_TH}})$ 1: $V(\text{VBRG}, \text{SB}) > (V_{\text{DS_TH}})$
BLM	2	Phase-B Low-side monitor 0: $V(\text{SB}, \text{LSS}) < (V_{\text{DS_TH}})$ 1: $V(\text{SB}, \text{LSS}) > (V_{\text{DS_TH}})$
AHM	1	Phase-A high-side monitor 0: $V(\text{VBRG}, \text{SA}) < (V_{\text{DS_TH}})$ 1: $V(\text{VBRG}, \text{SA}) > (V_{\text{DS_TH}})$
ALM	0	Phase-A Low-side monitor 0: $V(\text{SA}, \text{LSS}) < (V_{\text{DS_TH}})$ 1: $V(\text{SA}, \text{LSS}) > (V_{\text{DS_TH}})$

Diagnostic 1 Register

Reset value = 0x0000

GDU Address: 0x7E

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
			ENV	ESZ	EAC	ECRC	VC
RW	RW	RW	R/1C	R/1C	R/1C	R/1C	R/1C

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
VB	VA	CHO	CLO	BHO	BLO	AHO	ALO
R/1C	R/1C	R/1C	R/1C	R/1C	R/1C	R/1C	R/1C

Legend:

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
ENV	12	Serial Time Out If there a transaction is broken and no response from MCU then this flag is active. 0: Not active 1: Active
ESZ	11	Writing while NVM running. 0: Not active 1: Active If the I2C transaction is made while NVM is in active, this bit is flagged.
EAC	10	Serial Communication Acknowledge error 0: Not active 1: Active
ECRC	9	Serial CRC error 0: Not active 1: Active
VC	8	Phase C bootstrap undervoltage 0: Not active 1: Active
VB	7	Phase B bootstrap undervoltage 0: Not active 1: Active
VA	6	Phase A bootstrap undervoltage 0: Not active 1: Active
CHO	5	Phase C high-side VDS overvoltage 0: Not active 1: Active
CLO	4	Phase C low-side VDS overvoltage 0: Not active 1: Active
BHO	3	Phase B high-side VDS overvoltage 0: Not active 1: Active
BLO	2	Phase B low-side VDS overvoltage 0: Not active 1: Active
AHO	1	Phase A high-side VDS overvoltage 0: Not active 1: Active
ALO	0	Phase A low-side VDS overvoltage 0: Not active 1: Active

Status Register

Reset value = 0x7800

GDU Address: 0x07F

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
FF	POR	SE	EE	DSO	IG	OT	TW
R	R1C	R	R	R	R	R1C	R1C

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
BU	WD	ETO	SBOU	VCPU	VBU	VRU	VLRU
R	R1C	R1C	R1C	R1C	R1C	R1C	R1C

Legend:

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
FF	15	Status register flag 0: Non active 1: Fault detected If any of bit is flagged in the status register this bit also flag accordingly. Note: IG flag does not take effect to FF bit.
POR	14	Power-on-reset When the device first power on, this bit will be flagged.
SE	13	Serial Communication Error 0: Not active 1: Fault detected This bit is flagged when each of following bits active: ESTO, ENV, ESZ, EAC, ECRC
EE	12	NVM Error 0: Not active 1: Fault detected
DSO	11	VDS over voltage 0: Not active 1: Fault detected It is flagged when each of following bits active: AHO, ALO, BHO, BLO, CHO, CLO,
IG	10	Ignition Input state 0: Low (Less than IG low threshold) 1: High (Higher than IG high threshold)
OT	9	Over Temperature 0: Not active 1: Fault detected
TW	8	Temperature Warning 0: Not active 1: Fault detected
BU	7	Bootstrap Undervoltage 0: Not active 1: Fault detected It is flagged when each of following bits active: VC, VB, VA
WD	6	Q&A watchdog fault 0: Not active 1: Fault detected
ETO	5	ENABLE watchdog timeout 0: Not active 1: Fault detected
SBOU	4	Sensor BIAS terminal undervoltage 0: Not active 1: Fault detected
VCPU	3	VCP Undervoltage 0: Not active 1: Fault detected

VBU	2	VBB Undervoltage 0: Not active 1: Fault detected
VRU	1	VREG Undervoltage 0: Not active 1: Fault detected
VLRU	0	VLR Undervoltage 0: Not active 1: Fault detected

2.4 Supply and regulators

2.4.1 Main Power Supply

A single power supply voltage is required. The main power supply should be connected to VBB through a reverse voltage protection circuit. A 100nF ceramic decoupling capacitor must be connected close to the supply and ground terminals.

Four regulator circuits are included; a pump regulator for the gate drive, a pump regulator for above VBRG supply for external isolator, an external connected linear regulator to provide power for the MCU die, a regulator, which is for I/O interface reference voltage and core regulator.

The output voltage of the externally connected linear regulator, VLR, available at the VLR terminal, which generates 5V for MCU.

The core regulator is guaranteed to operate correctly to below the VBB POR level ensuring that the GDU will continue to operate safely until all logic is reset when a power-on-reset state is present.

The GDU will operate within specified parameters with VBB from 5.5V to 50V. Below 5.5V the gate drive outputs may be inactive but the GDU will continue to function with a supply down to 3.5V. It will remain in a safe state between 0V and 50V under all supply switching conditions. This provides a very rugged solution for use in the harsh automotive environment.

At power-up the logic inputs will be forced to their inactive state until VLR rises above the rising undervoltage threshold, VLRON.

If the WD mask bit is saved as 0 in non-volatile memory (NVM), the MRSTn output will remain low for tWI. If the WD mask bit is saved as 1 in NVM, the MRSTn output will remain low for tWI or until the first valid Q&A watchdog transfer (whichever occurs first). After the MRSTn output goes high the gate drive outputs will be re-enabled as described in the Fault Action section.

2.4.2 VLR Regulator

An integrated, programmable, linear regulator is provided to supply external logic level circuits such as a microcontroller or interface circuit. The output of the regulator on the VLR terminal is derived from VBB.

The regulator includes current limit, undervoltage and short protection. The current limiting circuit will reduce the output voltage to ensure that the output current does not exceed the current limit, ILROC. If the output voltage drops below the falling undervoltage threshold, V_{LROFF} , the MRSTn output will go low and can be used to reset an external microcontroller.

If the output voltage falls below the undervoltage detection, V_{LRUV_fal} , VLR undervoltage flag is reported to VLRU in the GDU status register. If the output voltage rises again above the rising threshold, V_{LRUV_ris} then the fault condition is cleared. But status flag will remain until cleared.

If the output voltage falls below the regulator shutdown threshold, V_{LROSD} , for a period exceeding the shutdown lockout period, 2[ms], the regulator is turned off and all logic inputs and outputs are disabled. In this state a small pilot current, 2 [mA] typically, is driven through the regulator output to detect load resistance. If the resultant voltage rises above the regulator shutdown threshold, V_{LROE} the regulator immediately attempts to restart.

At power-up, or when the regulator restarts, full output current is delivered for a period equal to the shutdown lockout period. During this time the output voltage is not monitored for short circuit conditions to ensure reliable regulator start-up.

If FOT = 1 in Stop on Fault register and the GDU internal junction temperature, T_J , rises above the overtemperature threshold, T_{JF} , the regulator is immediately shut down and MRSTn will go low. All GDU functions other than the regulator remains active. When T_J drops by more than the overtemperature hysteresis below the overtemperature threshold ($T_J < T_{JF} - T_{JHyst}$) the regulator will remain shut down and MRSTn will remain low for 10ms. After this timeout MRSTn goes high and the regulator is re-enabled and attempts to restart. If an undervoltage shutdown and an overtemperature warning occur simultaneously, both must be cleared to allow the regulator to restart.

Internal GDU logic circuitry is not powered from the VLR regulator and remains fully operational regardless of whether the VLR regulator is running normally or shut down.

A ceramic capacitor of at least 1 μ F with an ESR of no more than 250m Ω should be fitted between the VLR terminal and GND to guarantee stability and oscillation and voltage excursions outwith the specified output voltage range. In some applications the use of redundant output capacitors may be advisable to avoid such a condition in the event of a single-point capacitor-high-impedance failure.

2.4.3 VREG Charge Pump Regulator

The gate drivers are powered by a programmable voltage internal regulator which limits the supply to the drivers and therefore the maximum gate voltage. At low input supply voltage the regulated supply is maintained by a charge pump boost converter which requires a pump capacitor, typically 2.2 μ F, connected between the CP1 and CP2 terminals.

The regulated 11V, VREG, is available on the VREG terminal. A sufficiently large storage capacitor must be connected to this terminal to provide the transient charging current to the low side drivers and the bootstrap capacitors.

2.4.4 VCP Charge pump Regulator

This regulator generates a voltage above the bridge supply, VBRG. It provides some of the switching current for the high-side drives, the charge current for the bootstrap capacitors at high PWM duty cycles, the bias current and the gate drive current to hold the high-side MOSFET in the on state.

This regulator also uses a charge pump scheme with a switching frequency 62.5kHz. The voltage at the VREG terminal is pumped to produce the required output. Two external capacitors are required for the regulator to operate, one pump capacitor and one storage capacitor. The pump capacitor, which is between C3 and C4 terminal should have nominal value of 2.2 μ F. The storage capacitor, which is between VCP and VBRG should have a nominal value of 2.2 μ F. Both the pump and storage capacitors should have a rated working voltage of at least 25V and a tolerance of +20% or better.

2.4.5 Operating Modes

The GDU provides a sleep state and two operating modes related to the selected use of the LIN bus terminal; LIN mode and PWM mode. All configuration settings and basic control functions can be programmed through the serial interface in both LIN and PWM modes.

LIN Mode

When operating in LIN mode (OPM=0) the LIN terminal complies with the requirements of the LIN standard regarding dominant timeout and wake on LIN.

PWM Mode

When operating in PWM mode (OPM=1) the LIN dominant timeout is disabled and the wake mode is changed to wake on a simple rising or falling edge.

Low Power Sleep State

The GDU provides a low-power sleep state where the consumption from the supply is reduced to a minimum by disabling all normal operation including the charge pump regulator, the internal logic regulator, the VLR regulator and the internal clock. In the sleep state the LIN terminal must be at the same voltage as the supply terminal, VBB, and the IG terminal should be tied to ground to achieve the minimum supply current.

When operating in LIN mode (OPM=0) the GDU can only be commanded to go into the sleep state using the serial interface to change the GTS bit from 0 to 1 when the LIN input is high (recessive). If the LIN input is low (dominant) any sleep command through the serial interface will be ignored and the GTS bit will have to be changed to 0 followed by 1 to issue another go-to-sleep command when the LIN input is high (recessive). When operating in PWM mode (OPM=1) the GDU will go into the normal sleep state using the serial interface to change the GTS bit from 0 to 1, irrespective of the level on the LIN terminal.

The sequence to wake the GDU is determined by the operating mode defined by the OPM bit. When LIN mode is selected (OPM=0) the GDU will wake up according to the LIN protocol. When the PWM mode is selected (OPM=1) the GDU will wake up on any valid transition of the signal at the LIN terminal. These sequences are fully described and defined in the LIN interface section.

In either operating mode the GDU will also wake up on a low to high transition of the signal on the IG terminal.

In the sleep state the MRSTn output is held low, latched faults are cleared and the Diagnostic and Status registers are reset to zero.

When coming out of the sleep state all registers are reset to default state.

Table 1: Operating and wake mode

	Operating Mode	
	LIN	PWM
OPM	0	1
Sleep (I2C) Command	I2C (GTS 0->1) [Only when LIN is high]	I2C (GTS 0->1)
Sleep (IG) Command	No	No
Wake (IG)	IG L->H	IG L->H
Wake (LIN)	First L->H transition on LIN terminal after LIN terminal low for $>t_{BUSWK}$	Any transition on LIN terminal present for $>t_{BUSWK}$

The user defined values held in the non-volatile memory and the GDU follows the same procedure as for a full poweron reset. MRSTn is held low until t_{WI} after the VLR regulator output exceeds its undervoltage threshold. In addition, the charge pump output monitor ensures that the gate drive outputs are off until the charge pump reaches its correct operating condition. The charge pump will stabilize in approximately 2ms under nominal conditions.

Microcontroller Reset & Watchdog

The microcontroller reset output, MRSTn, can be used to reset and re-initialize an external microcontroller if an undervoltage, watchdog fault or power-on-reset (POR) occurs. The MRSTn output will be active low when the external regulator undervoltage or POR fault state is present and will remain low for XXms after all faults are removed. It will also go low for XXms when a watchdog fault is detected.

Q&A Watchdog

The GDU includes a programmable Q&A watchdog that can be used to determine if the MCU is operating in an adverse state. The Q&A watchdog provides time out diagnostics to detect the MCU is no longer responsive. When Q&A watchdog is in active, WD=0 on Mask0, the valid Q&A transaction reset the watchdog timer. The MCU is required to send consecutive valid commands to keep active the gate driving function. If GDU detects Q&A watchdog fault, then it automatically disable the bridge to be safe state on GDU.

First of all to make a successful Q&A watchdog transaction, the master is required to read back Q&A watchdog lookup table address (Question). The Q&A watchdog look-up table address can be read out via QA[15:0] in Q&A watchdog register. When the master confirms the address, it is required to send back correct contents (Answer) into QA[15:0] register. When the successful answer is accepted, then Q&A watchdog timer will be reset and remain the bridge active. When an answer is received regardless the answer is incorrect or not, the lookup table address increment 1 for the next transaction and the master is required to send out correct answer within timeout period, t_{WO} . The lookup table is described in GDU register section.

When the GDU at power on or come out from sleep state, and if the Q&A watchdog is active, the Q&A watchdog timer is waiting first transition for an extended time. This scheme allows the MCU carries out power-on initialization. The power-on watchdog time-out period is defined as $t_{WI} + t_{WO}$, which is selectable by WI and WO. If the GDU detects a Q&A watchdog error, then MRSTn is toggled as low to send reset signal to MCU. MRSTn signal holds low for t_{WI} and set back to high. The GDU then wait power-on watchdog time-out period until GDU receives a valid Q&A watchdog frame. The WD in status register will be hold until cleared.

After the Q&A watchdog is active, MCU requires to send valid Q&A watchdog frame repeatedly within t_{WO} , which is selected by WO register. When the Q&A watchdog receives valid frame, GDU reset the watchdog counter then wait next valid frame. If the GDU detects a Q&A watchdog error, then MRSTn is toggled as low to send reset signal to MCU. MRSTn signal hold low for t_{WI} and set back to high. The GDU then wait power-

on watchdog time-out period until GDU receives a valid Q&A watchdog frame. The WD in status register will be hold until cleared.

When a watchdog failure is detected the gate drive outputs are disabled (low) and any attached motor will coast. The gate drive outputs remain disabled until a valid Q&A watchdog transition is detected. Once a valid watchdog has been detected the GDU will re-enable the gate drive outputs.

If the watchdog function is not required it is possible to disable the function by setting the WD bit in the mask register to 1. This will completely disable the Q&A watchdog monitor and any possible actions that it may take.

If the microcontroller has completely stopped working it is possible to put the GDU into the sleep state, after a number of watchdog time-out period using the watchdog fail cycle count variable, WC[3:0]. The value in WC[3:0] sets the number of watchdog fail – reset cycles that can occur before the GDU goes into the sleep state. A value of 1 will allow one fail-reset cycle and will go to sleep on the next watchdog failure if no valid transitions are detected. A value of 8 will allow 8 fail-reset cycle and will go to sleep on the 9th watchdog failure if no transitions are detected. The counter is reset if any valid transition is detected. A value of zero in WC[3:0] will disable this feature and permit unlimited fail-reset cycles.

The sequence to wake the GDU from the sleep state is described in the Low Power Sleep State section.

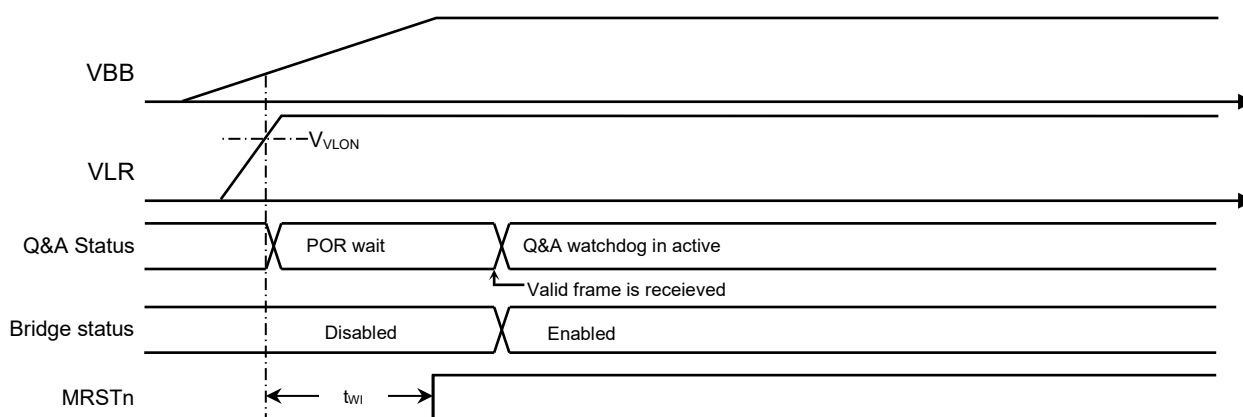


Figure 3a: Q&A watchdog activation after power-on

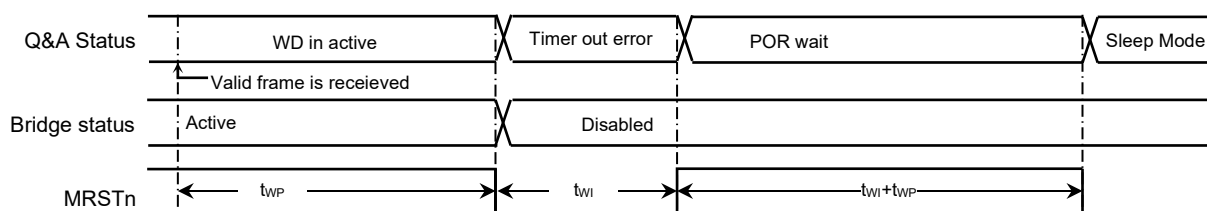


Figure 3c: Sleep after fail cycle

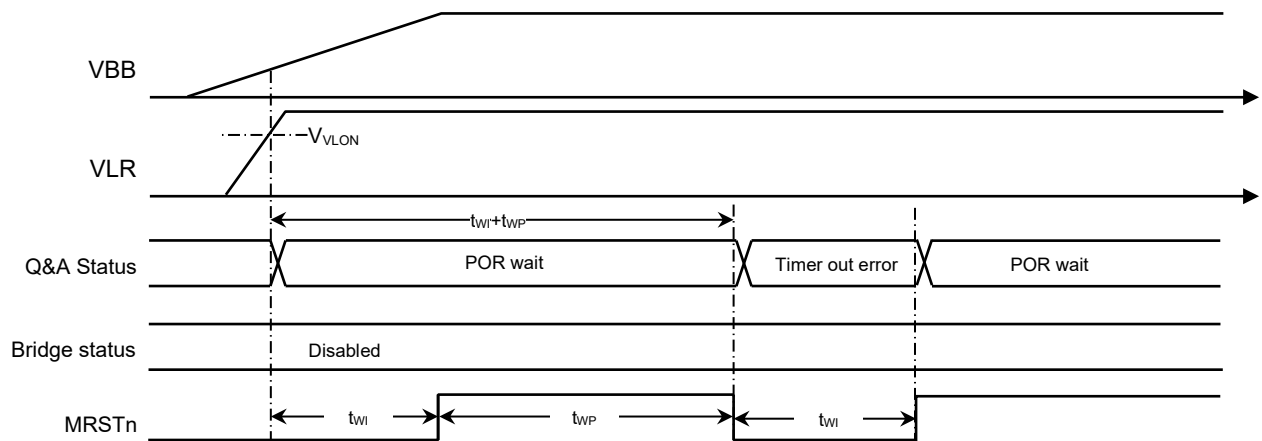


Figure 3b: Q&A watchdog activation failure after power-on

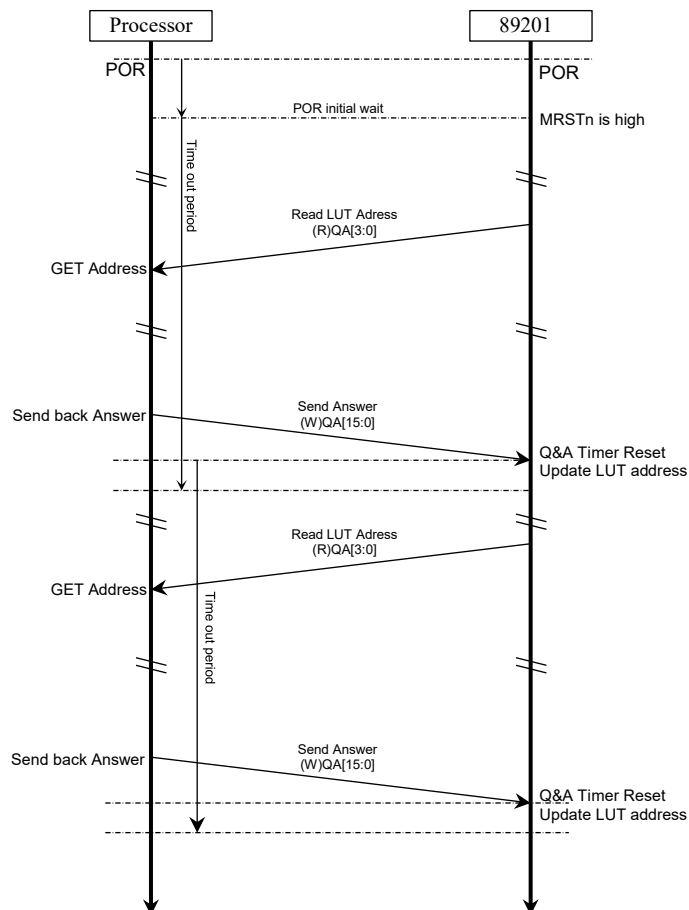


Figure 4: Q&A Communication Sequence

LIN Physical Interface

The GDU includes a physical interface to drive and monitor a single wire LIN Bus as a slave node that complies with the ISO17987 (LIN 2.x) and SAE J2602 standards. The LIN terminal can withstand voltages from -40V to +50V with respect to the ground pin without adversely affecting LIN Bus communications between other devices.

The LIN terminal meets all the voltage, timing and slew limitation requirements of LIN 2.2 when actively transmitting, and when receiving. When operating in LIN mode (OPM=0), a timer is included to ensure that the LIN output does not remain dominant when a fault occurs. If the LIN terminal is driven in the dominant state for longer than the transmit dominant timeout period, t_{TXTO} , then the output is disabled and allowed to return to the recessive state in order to avoid locking the LIN Bus for other messages. The dominant timeout function is disabled in the PWM operating mode (OPM=1) or when the LIN interface is in the standby state (LEN=0).

The data to be transmitted is input to the LTX terminal and converted to LIN Bus signals. A high on the LTX produces a recessive bus (high) state while a logic low produces a dominant bus (low) state. The LTX input has an internal pullup resistor to ensure a recessive state if the pin is not connected or becomes disconnected.

The logic state of the LIN Bus is determined by the receiver and output as a logic level on the LRX terminal. LRX will be low when the LIN Bus is in the dominant (low) state and high when the LIN Bus is in the recessive (high) state. In the sleep state LRX is not active and will be low.

In PWM mode the LIN terminal can also be used as a PWM interface to the external microcontroller. The level of the PWM signal applied to the LIN terminal will be output as a logic level on the LRX terminal. When operating in PWM mode the LTX input can be used to pull the PWM signal low in order to indicate a fault to the external ECU.

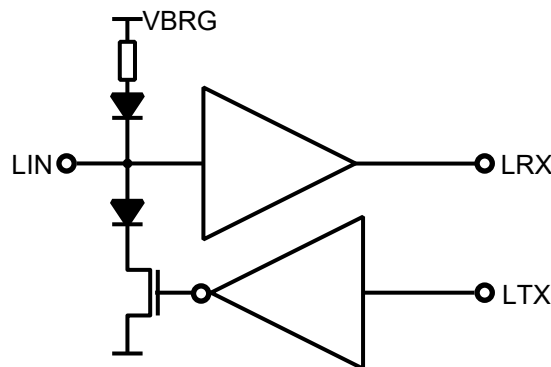


Figure 5: LIN Physical Interface

When the GDU is in the sleep state the LIN terminal changes to a passive input and the resistance of the pull-up resistor on the LIN terminal increases to approximately 2MΩ. This ensures that the LIN terminal is unable to affect the LIN bus signal. When DWL = 0, the LIN terminal continues to be monitored in the normal sleep state in order to detect a wake request.

Two wake sequence modes are possible, selected by the operating mode.

In LIN mode (OPM=0) the GDU will wake up according to the LIN protocol. In this mode, the wake request is valid on the first low-to-high transition on the LIN terminal after the LIN terminal is in a dominant (low) state for longer than the wake time, t_{BUSWK} , as shown in figure 6a. If the LIN terminal changes to recessive (high) within t_{BUSWK} then the GDU returns to the sleep state.

In PWM mode (OPM=1) the GDU will wake up on any valid transition of the signal at the LIN terminal. In this mode, the wake signal is valid when the signal on the LIN terminal changes state, high to low or low-to-high, and

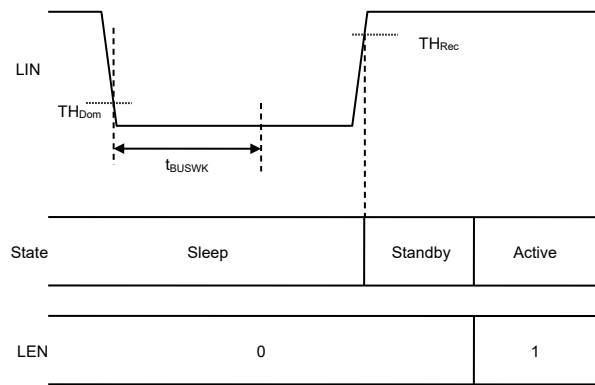


Figure 6a: LIN Wake Sequence & Timing

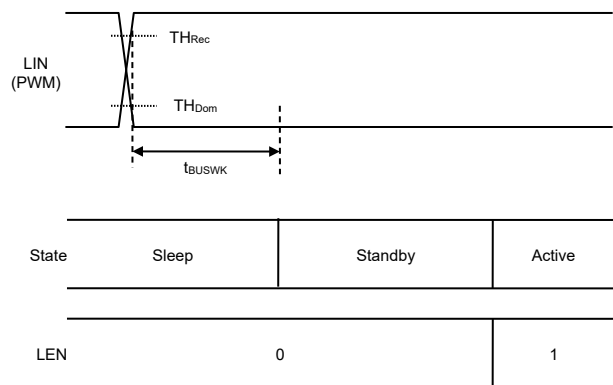


Figure 6b: PWM Wake Sequence & Timing

remains in the changed state for longer than the wake time, t_{BUSWK} , as shown in figure 6b. If the LIN terminal does not remain in the changed state for longer than the wake time, t_{BUSWK} , then the GDU returns to the sleep state.

When a valid wake request is detected on the LIN terminal, with $DWL = 0$, or the IG terminal transitions from low to high the GDU will exit normal the sleep state, turn on all regulators and control circuits and commence operation.

At this time, if the LEN bit is 0, the LIN interface will remain in the standby state where the LIN terminal is a passive input. The LRX output will indicate the state of the LIN terminal but the signal on the LTX terminal is ignored. The LIN interface becomes fully active when LEN is set to 1. If the default value of LEN is 1 then the standby state is bypassed and the LIN interface is fully active as soon as the internal regulators are fully active.

There are LIN terminal slew-rate control. When the LBR bit is set 0, LIN terminal slew-time is optimized with 10kHz transmission. When the LBR is set 1, LIN terminal slew-time is optimized with 20kHz. This function allows to optimize EMC performance.

MCU Connection:

LTX and LRX are internally connected to the MCU, and the signal can be configured in GPIO section. Please refer GPIO manual to use LIN physical interface.

Gate Drive Voltage Regulation

The gate drivers are powered by an integrated programmable voltage regulator which limits the supply to the drivers and therefore the maximum gate voltage. At low supply voltage the regulated supply is maintained by a charge pump boost converter which requires a pump capacitor, typically 2.2μF, connected between the CP1 and CP2 terminals.

The regulated voltage, VREG, is 11V and is available on the VREG terminal. A sufficiently large storage capacitor must be connected to this terminal to provide the transient charging current to the low side drivers and the bootstrap capacitors.

Bootstrap Supply

When a high-side driver is active the reference voltage, S_x , will rise to close to the bridge supply voltage. The supply to the driver will then have to be above the bridge supply voltage to ensure that the driver remains active. This temporary high-side supply is provided by bootstrap capacitors, one for each high-side driver. These three bootstrap capacitors are connected between the bootstrap supply terminals, CA, CB, CC, and the corresponding highside reference terminal, SA, SB, SC.

The bootstrap capacitors are independently charged to approximately VREG when the associated reference terminal, S_x , is low. When the output swings high, the voltage on the bootstrap supply terminal, C_x , rises with the output to provide the boosted gate voltage needed for the high-side nchannel power MOSFETs.

Bootstrap Charge Management.

The GDU monitors the individual bootstrap capacitor charge voltages to ensure sufficient high-side drive.

Before a high-side drive can be turned on, the bootstrap capacitor voltage must be higher than the turn-on voltage threshold, $V_{BCUV} + V_{BCUVHys}$. When any of three high-side are commanded, bootstrap undervoltage monitor will be active accordingly. If the voltage is below undervoltage threshold, V_{BCUV} , undervoltage will be detected when it is commanded.

When the commanded PWM duty cycle is very high, there may not be enough charge from turning on corresponding low-side MOSFET. When the duty cycle is very high, the bootstrap capacitor will be charged from VCP supply. This allows having 0% to 100% duty cycle command without no restriction. When the VCPM is 0~2, the VCP power supply is available to top-up the bootstrap capacitor. When the VCPM is 0, automatic mode will be apply. When any of high-side is commanded to turn-on, high-current, 2.33mA will be applied for 200[μs] period. After that, the VCP current is limited to 200μA. This scheme makes sure the bootstrap voltage at high-duty cycle and optimize the power loss when it is fully turned on. When the VCPM is 1 or 2, the current level is fixed. Fixed high curent, VCPM=2 is useful to connect an external circuit above battery supply. When the VCE is 3, VCP chargepump will be turned off and there will be some limitation when the commanded duty cycle is high.

High-side gate drive

High-side, gate-drive outputs for external n-channel power MOSFETs are provided on terminals GHA, GHB and GHC.

GHx=1 (or “high”) means that the upper half of the driver is turned on and its drain will source current to the gate of the high-side MOSFET in the external motor-driving bridge, turning it on. GHx=0 (or “low”) means that the lower half of the driver is turned on and its drain will sink current from the external MOSFET’s gate circuit to the respective S_x terminal, turning it off.

The reference points for the high-side drives are the load phase connections, SA, SB and SC. These terminals sense the voltages at the load connections. These terminals are also connected to the negative side of the bootstrap capacitors and are the negative supply reference connections for the floating high-side drivers. The discharge current from the high-side MOSFET gate capacitance flows through these connections which should have low impedance traces to the MOSFET bridge.

MOSFET gate charge and discharge rates may be controlled via the serial interface as detailed in the Gate Drive Control section.

Low-side gate drive

The low-side, gate-drive outputs on GLA, GLB and GLC are referenced to the LSS terminal. These outputs are designed to drive external n-channel power MOSFETs. GLx=1 (or “high”) means that the upper half of the driver is turned on and its drain will source current to the gate of the low-side MOSFET in the external power bridge, turning it on. GLx=0 (or “low”) means that the lower half of the driver is turned on and its drain will sink current from the external MOSFET’s gate circuit to the LSS terminal, turning it off.

The LSS terminals provide the return paths for discharge of the capacitance on the low-side MOSFET gates. These terminals are connected independently to the common sources of the low-side external MOSFETs through a low impedance tracks.

MOSFET gate charge and discharge rates may be controlled via the serial interface as detailed in the Gate Drive Control section below.

Gate drive passive pull-down.

Each gate drive output includes a discharge circuit to ensure that any external MOSFET connected to the gate drive output is held off when the power is removed. This discharge circuit appears as a variable resistance pull-down, but is not active when the GDU is in normal operating mode. At low gate source voltage the resistance is approximately 950k Ω to ensure that any charge accumulated on the MOSFET gate has a discharge path. This resistance reduces rapidly as the voltage increases such that any MOSFET gate that becomes charged by external means is rapidly discharged to below the turn-on threshold. In some applications this can eliminate the requirement for a permanent external gate source resistor.

Gate drive control

In some applications it may be necessary to limit the rate of change of the voltage at the motor phase connections to help comply with EMC emission requirements. This is usually achieved by controlling the MOSFET gate charge and discharge rates.

The conventional approach is to add an external resistor between the gate drive output and the gate connection to each MOSFET and possibly an additional small value capacitor between the gate and source of the external MOSFET.

In addition to operating in this basic switch mode drive, the GDU gate drive output can be programmed to provide control of the slew rate of the drain-source voltage of the external power MOSFET. This is achieved by controlling the gate sink or source current during the time when the drain-source voltage is changing. This occurs during the Miller region when the gate-drain capacitance of the external MOSFET is being charged or discharged. This capacitance is referred to as the Miller capacitor and the period of time as the Miller time. MOSFET gate drives are controlled according to the values set in the slew control variables IR1, IR2, IF1, IF2, TRS and TFS. The off-to-on transition is controlled by IR1, IR2, and TRS. The on-to-off transition is controlled by IF1, IF2, and TFS.

There are two gate drive control modes, switched and slew control. All gate drives operate in the same mode.

In switched mode the gates are driven at the full capability of the pull-up or pull-down switches in the gate drive, as shown in figure 7b. If both IR1 and IR2 are set to zero the gate drive operates in full switched mode for the off-to-on transition. If both IF1 and IF2 are set to zero the gate drive operates in full switched mode for the on-to-off transition. If the dead time is set to zero by setting the DT variable to zero then the gate drive can only operate in switched mode.

In slew-control mode the gates are driven using programmable currents to provide some control over the slew rate of the motor phase connection as shown in figure 7a. To operate in slew-control mode for the off-to-on transition both IR1 and IR2 must be non-zero. To operate in slew-control mode for the on-to-off transition both IF1 and IF2 must be non-zero. If any of the drive currents are set to zero then the output will operate in switched mode for the period of time when that current is active.

The basic principle of the slew rate mode is to drive the gate with a controlled current for a fixed time to quickly get the MOSFET to the Miller region where the drain-source voltage, VDS, starts to change. Then to follow this with a second, usually lower current to control the VDS slew rate. Finally the gate drive changes to switch mode once VDS has completed its transition.

In slew control mode, when a gate drive is commanded to turn on, a current, IR1 (defined by IR1[3:0]) is sourced from the relevant gate drive output for a duration, tRS (defined by TRS[3:0]). These parameters should typically be set to quickly charge the MOSFET input capacitance such that the gate-source voltage rises close to the Miller voltage of the MOSFET. The drain-source voltage of the MOSFET will not start to change until the gate-source voltage reaches this level. After this time, the current sourced on the gate drive output is set to a value of IR2 (as defined by IR2[3:0]) and remains at this value while the MOSFET transitions through the Miller region. IR2 should be selected to achieve the required slew rate of the drain-source voltage by setting the charge time of the drain-gate (Miller) capacitor.

When the MOSFET reaches the fully-on state the gate drive output changes from current drive to voltage drive to hold the MOSFET in the on state.

A high-side MOSFET is considered to be in the fully-on state when the drain-source voltage, V_{DS} ($=V_{BB}-V_{Sx}$), drops below the programmed V_{DS} threshold voltage, V_{DST} .

A low-side MOSFET is considered to be in the fully-on state when the drain-source voltage, V_{DS} ($=V_{Sx}-V_{GND}$), drops below the programmed V_{DS} threshold voltage, V_{DST} .

When a gate drive is commanded to turn off in slew control mode, a current, I_{F1} (defined by $I_{F1}[3:0]$) is sunk to the relevant gate drive output for a duration, t_{FS} (defined by $T_{FS}[3:0]$). These parameters should typically be set to quickly discharge the MOSFET input capacitance such that the gate-source voltage drops to close to the Miller voltage of the MOSFET. The drain-source voltage of the MOSFET will not start to change until the gate-source voltage reaches this level. After this time, the current sourced on the gate drive output is set to a value of I_{F2} (as defined by $I_{F2}[3:0]$) and remains at this value while the MOSFET transitions through the Miller region. I_{F2} should be selected to achieve the required slew rate of the drain-source voltage by setting the discharge time of the drain-gate (Miller) capacitor.

When the MOSFET reaches the fully-off state the gate drive output changes from current drive to voltage drive to hold the MOSFET in the off state.

A high-side MOSFET is considered to be in the fully-off state when the drain-source voltage of its complementary low-side MOSFET, V_{DS} ($=V_{Sx}-V_{GND}$), drops below the programmed V_{DS} threshold voltage, V_{DST} .

A low-side MOSFET is considered to be in the fully-off state when the drain-source voltage of its complementary highside MOSFET, V_{DS} ($=V_{BB}-V_{Sx}$), drops below the programmed V_{DS} threshold voltage, V_{DST} .

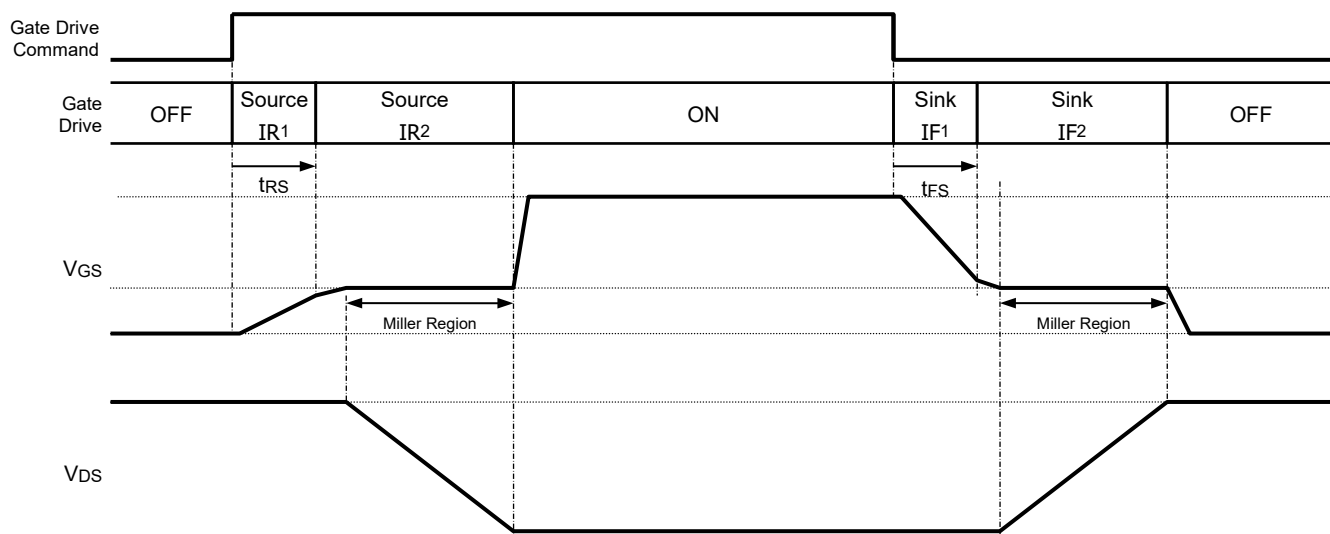


Figure 7a: Off-to-On and On-to-Off Transitions with Gate Drive Control enabled

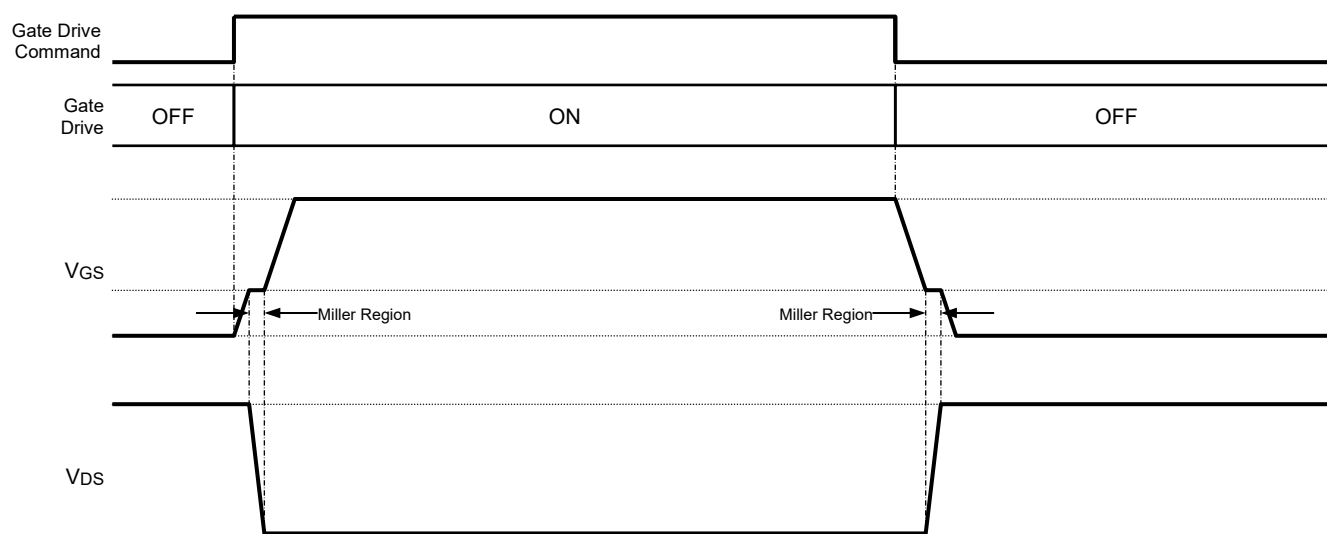


Figure 7b: Off-to-On and On-to-Off Transitions with Gate Drive Control Disabled

Logic Control Inputs

Six logic level digital inputs, HA, LA, HB, LB, HC, LC, provide direct control for the gate drives, one for each drive. The Hx inputs correspond to the high-side drives and the Lx inputs correspond to the low-side drives. The six control block are internally connected as follows:

MCU Terminal	GDU Terminal
GPIO10 (PD10)	HA
GPIO11 (PD11)	LA
GPIO12 (PD12)	HB
GPIO13 (PD13)	LB
GPIO14 (PD14)	HC
GPIO15 (PD15)	LC

Note: Before use 6-PWM signal properly, GPIO need to be configured. Please refer the GPIO section for more configuration detail.

The operation of the inputs are shown in the Control Logic Table, table 1.

The gate drive outputs can also be controlled through the serial interface by setting the appropriate bit in the control register. In the control register all bits are active high. The logical relationship between the register bit setting and the gate drive outputs is defined in table 3.

The logic inputs are combined, using logical OR, with the corresponding bits in the serial interface control register to determine the state of the gate drive. The logical relationship between the combination of logic input and register bit setting and the gate drive outputs is defined in table 4. In most applications either the logic inputs or the serial control will be used. When using only the logic inputs to control the bridge the serial register should be left in the reset condition with all control bits set to 0. When using only the serial interface to control the bridge the inputs should be set as 0.

Internal lock-out logic ensures that the high-side gate drive output and low-side gate drive output drive on the same phase cannot be active (high) simultaneously. If the control inputs request both the high-side and the low-side gate drives to be active at the same time then both high-side and lowside gate drives are switched into the inactive (low) state.

Table2: Control logic table: Control by GPIO

Phase A					Phase B					Phase C				
HA	LA	GHA	GLA	SA	HB	LB	GHB	GLB	SB	HC	LC	GHC	GLC	SC
0	0	LO	LO	Z	0	0	LO	LO	Z	0	0	LO	LO	Z
0	1	LO	HI	LO	0	1	LO	HI	LO	0	1	LO	HI	LO
1	0	HI	LO	HI	1	0	HI	LO	HI	1	0	HI	LO	HI
1	1	LO	LO	Z	1	1	LO	LO	Z	1	1	LO	LO	Z

Note: HI: High-side FET active, LO: Low-side FET active

Z: High impedance, both FETs off

All control Register bits set to 0, gate driver outputs active.

Table3: Control logic table: Control by Serial register

Phase A					Phase B					Phase C				
AH	AL	GHA	GLA	SA	BH	BL	GHB	GLB	SB	CH	CL	GHC	GLC	SC
0	0	LO	LO	Z	0	0	LO	LO	Z	0	0	LO	LO	Z
0	1	LO	HI	LO	0	1	LO	HI	LO	0	1	LO	HI	LO
1	0	HI	LO	HI	1	0	HI	LO	HI	1	0	HI	LO	HI
1	1	LO	LO	Z	1	1	LO	LO	Z	1	1	LO	LO	Z

Note: HI: High-side FET active, LO: Low-side FET active

Z: High impedance, both FETs off

Logic 0 input on HA, LA, HB, LB, HC, LC though GPIO. Gate drive output active.

Table 4: Control combination logic table: Logic inputs and Serial Register

GPIO	Register	Internal	GPIO	Register	Internal
Hx	xH	Hlx	Lx	xL	LOx
0	0	0	0	0	0
0	1	1	0	1	1
1	0	1	1	0	1
1	1	1	1	1	1

Hlx	LOx	GHx	GLx	Sx	Comment
0	0	LO	LO	Z	Phase Disabled
0	1	LO	HI	LO	Phase sinking
1	0	HI	LO	HI	Phase sourcing
1	1	LO	LO	Z	Phase disabled

The three phases are controlled independently.

Internal Control signals (Hlx, LOx) are derived by combining the logic state applied to the control input terminals (Hx, Lx) with the bit patterns held in the Control register (xH, xL).

BEMF & Phase Voltage Output

The voltage on the Sx terminal for each phase is compared to a common “centre tap” voltage, VCT, generated by the average of the three phase voltages defined as:

$VCT = (VSA + VSB + VSC)/3$ The signal output on the CMPA, CMPB, CMPC terminal is selected by the CPM[2:0] variable as shown in table 5.

Table 5: Comparator output select

CPM2	CPM1	CPM0	CMPA	CMPB	CMPC
0	0	0	A-phase BEMF	B-phase BEMF	C-phase BEMF
0	0	1	A-phase comp	B-phase comp	C-phase comp
0	1	0	Low	Low	Low
0	1	1	High	High	High
1	X	X	Low	Low	Low

Figure 8 provides a functional diagram and figure 9 shows the output options for a normal motor rotation when the gate drive outputs are disabled and the motor is not being driven. There are phase comparator selection on CMPx output. If the phase comparator is chosen, the Sx node is compared with programable threshold, VPT. The threshold can be configured by VPT[5:0]. The parametric performance is described in the electrical characteristics table.

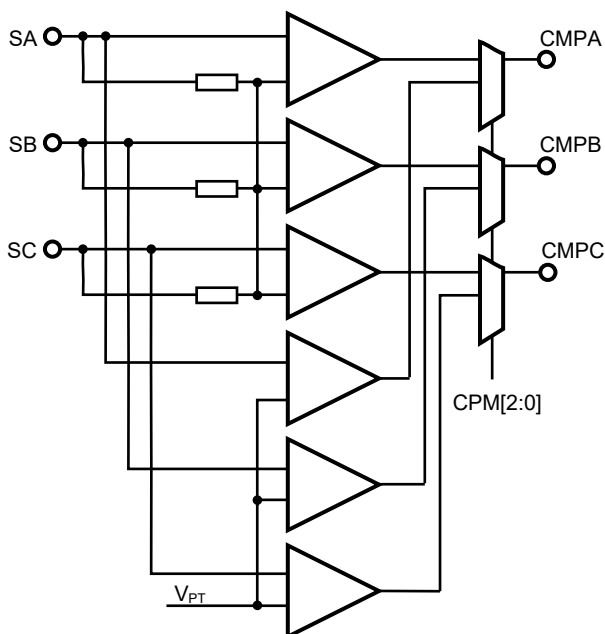


Figure 8: BEMF Detection Multiplexer Diagram

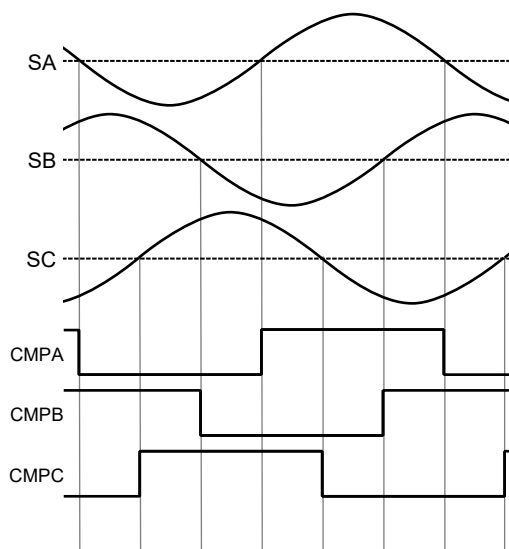


Figure 9a: Multiplexer Output Selection (CPM = 0)

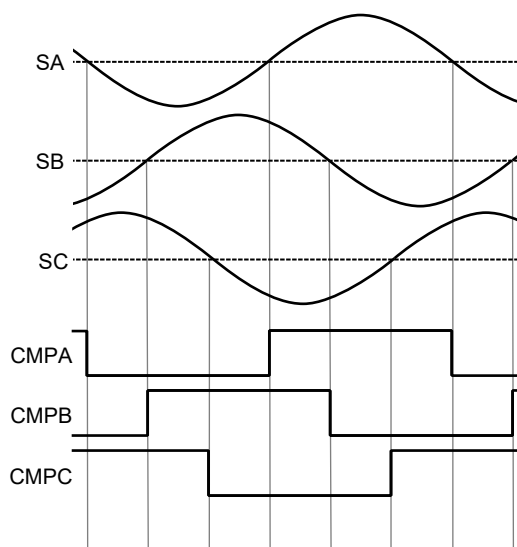


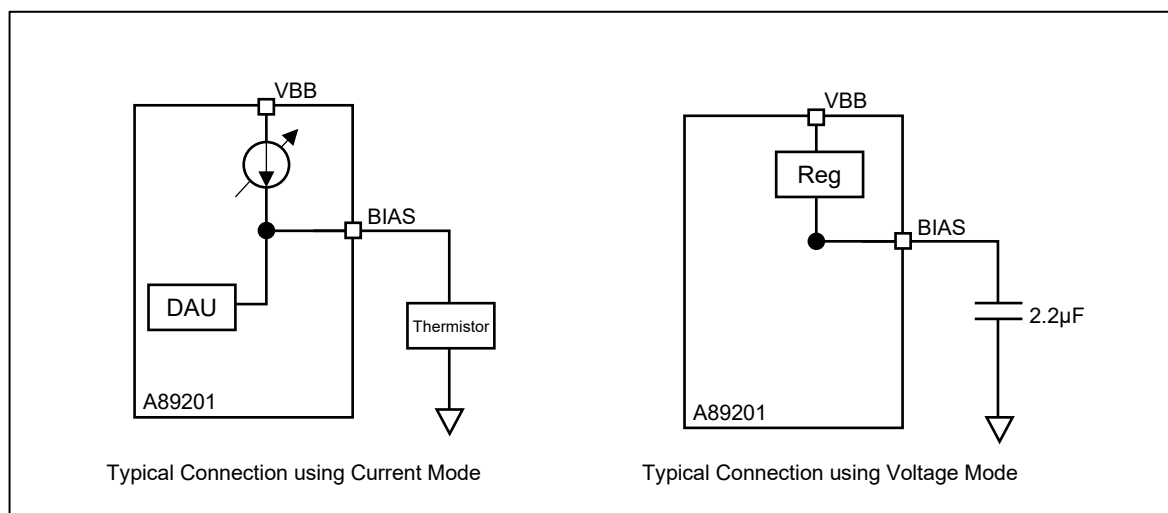
Figure 9b: BEMF Multiplexer Output Selection (CPM = 0)

Sensor BIAS Terminal

The GDU has sensor bias terminal, BIAS, which is battery voltage compliant. The BIAS terminal allows to provide a fixed current or fixed voltage for external purpose. This allows to use external thermister to measure a temperature effectively with current mode. The voltage mode can be used for external hall-effect sensor for example.

The sensor BIAS mode can be chosen via SBM[2:0] register. When SBM = 0, the BIAS terminal is disabled, the terminal will be high-impedance terminal. When SBM = 4 or 5, the BIAS terminal will behave as constant current output mode. The current level can be chosen by SBC[2:0]. The current level can be chosen from 25 μ A to 2.5mA. The device will force the current until the terminal voltage gets satulated.

When the SBM = 6 or 7, the BIAS terminal will behave as constant voltage mode. 2.2 μ F ceramic capacitor has to be connected to the BIAS terminal to stablize the voltage. The output voltage is specified in the datasheet.



In current mode it is possible to measure the BIAS terminal voltage and two settings are required. Firstly, AMX[3:0] setting need to be configured as BIASRB, 0xB. When it is selected, voltage is available for DAW. Secondly, DAW setting need to be configured, please refer the DAW section more detail.

Ignition terminal (IG)

The GDU has an ignition terminal, which has battery compliant pull-down circuit. Ignition terminal can be used to wake-up from the sleep mode, and the state of IG terminal can be read out via IG on status Register. For more detail of wake-up feature, please refer the Low-Power Sleep state section.

Fault reporting terminal (FAULTn)

The GDU has FAULTn terminal, which has battery compliant open-drain circuit. The open-drain transistor can be controlled via FLT in the gate driver control Register. When FLT = 0, the open-drain transistor is off, so the terminal will be high impedance. When FLT = 1, the open-drain transistor is on, so the FAULTn terminal pull to GND.

Current Sense Differential Amplifier

A single programmable-gain differential sense amplifier is provided to allow the use of a low value sense resistor or current shunt as a current sensing element in the connection between the common source connections of the low-side MOSFETS to ground. The input common mode range of the CSP and CSM inputs allows below ground current sensing typically required for low-side current sense in PWM control of motors, or other inductive loads, during switching transients. The output of the sense amplifier is available at the CSO output and can be used in peak or average current control systems. Dedicated differential outputs are internally connected to MCU's CAU unit. This allows to have accurate current sampling.

The differential amplifier's gain is programmable and defined by the contents of the DAG[2:0] variable as:

DAG	Gain	DAG	Gain
0	5	4	15
1	8	5	18
2	10	6	20
3	13	7	25

Current Sense Single Ended Amplifier (Only Available for option A and C)

A89201 has a single ended amplifier. This option is only available option A and C. The input of single ended amplifier is from differential inputs, CSP and CSM. It allows to sense the bi-directional current into single ended output. This structure typically is used for measuring average bus current.

The gain, AV, of the sense amplifier is programmable and defined by the contents of the SG[2:0] variable as:

SG	Gain	SG	Gain
0	10	4	30

1	16	5	36
2	20	6	40
3	26	7	50

The output offset of the sense amplifier, VOOS, is defined by the contents of the SAO[3:0] variable as:

SAO	VOs	SAO	VOs
0	0	8	720mV
1	0	9	960mV
2	100mV	10	1.2V
3	100mV	11	1.59V
4	200mV	12	1.65V
5	300mV	13	1.92V
6	400mV	14	2.16V
7	500mV	15	2.4V

Equations describing the relationship between sense amplifier gain and output offset are shown in figure 10.

To achieve the sense amplifier Input Offset Voltage, VIOS, limits detailed in the Electrical Characteristics table the amplifier must be calibrated and output offset uncertainty must be eliminated.

The current sense amplifier calibration is initiated via the serial interface. Amplifier is calibrated by writing a 1 to the SC bit. If the SC bit is already set to 1 it must first be cleared to 0 before writing 1 otherwise a calibration cycle will not take place. Before initiating a calibration, the OSO bit must be set to 0 and the relevant positive and negative amplifier input terminals (CSP and CSM) must be held at the same potential by ensuring that no current is flowing in the sense resistor. This condition must be maintained until the calibration operation is complete. Calibration starts on the STRn rising edge associated with writing 1 to the SC bit and is completed within an Offset Calibration Time, tCal, from this point. After a calibration the amplifier automatically reverts to normal operating mode. During calibration transient voltage variations may be observed on the CSO terminal.

Calibration is required to achieve the specified Input Offset Voltage Drift, ΔVIOS, and leaves this parameter unaltered.

Output offset uncertainty is eliminated by carrying out a two-stage voltage measurement on the CSO terminal. Firstly, the OSO bit is set to 0 to make the combined current sense signal plus output offset available on

CSO and this is measured as $VCSO(1) = VCSD + VOOS$. Secondly, the OSO bit is set to 1 to make only the output offset voltage available on CSO and this is measured as $VCSO(2) = VOOS$. The actual current sense signal is then determined by calculating the difference between the two measurements as follows:

$$VCSO(1) - VCSO(2) = (VCSD + VOOS) - VOOS$$

$$VCSO(1) - VCSO(2) = VCSD$$

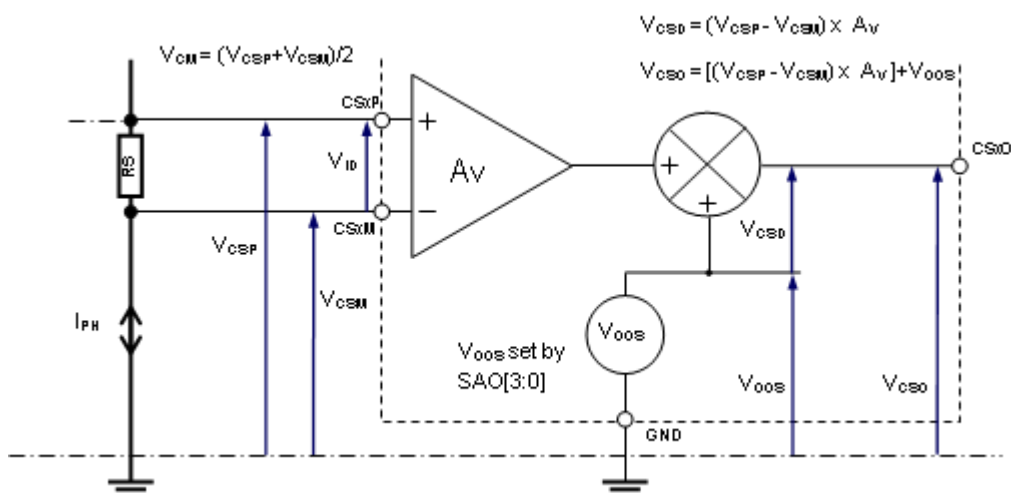


Figure 10: Sense amp voltage definitions

Analog Multiplexer Output

The GDU analog multiplexer output provides access to analog voltages on GDU. Analog mux output can be access by DAU. The output can be selected by AMX from VBB, VBRG, VSA, VSB, VSC, VREG, VCP, VLR, IG and Tj. The full-scale output voltage range at AOUT is set at 2.4V to match the input to the ADC. The division ratio for each input is shown in electrical characteristics table on the datasheet and the node selection by AMX is shown in table 6.

AMX				AOUT
3	2	1	0	
0	0	0	0	VBB
0	0	0	1	VBRG
0	0	1	0	SA
0	0	1	1	SB
0	1	0	0	SC
0	1	0	1	VREG
0	1	1	0	VCP
0	1	1	1	VLR
1	0	0	0	Tj
1	0	0	1	IG
1	0	1	0	OOS
1	0	1	1	BIAS
1	1	0	0	Unused
1	1	0	1	GND
1	1	1	0	GND
1	1	1	1	GND

Diagnostics

Multiple diagnostic features provide fault monitoring for the basic chip status and for the state of the external bridge. Most of the non-critical diagnostics can be masked by setting the appropriate bit in the mask register. The detectable faults are listed in table 7. The fault status is available through the serial interface.

Status register

The serial interface allows detailed diagnostic information to be read from the Status register. Table 7: Diagnostic Functions

Name	Diagnostic	Level
------	------------	-------

POR	Internal logic supply undervoltage causing power-on reset	Chip
SE	Serial transmission error	Chip
EE	NVM error	Chip
DSO	General VDS overvoltage	Bridge
IG	IG terminal Status	Terminal
OT	Chip junction over temperature	Chip
TW	High chip junction temperature warning	Monitor
BU	Bootstrap Undervoltage	Monitor
WD	Q&A watchdog	Monitor
ETO	ENABLE watchdog timeout	Monitor
SBOU	Sensor BIAS Terminal Undervoltage	Monitor
VCPU	VCP output undervoltage	Monitor
VBU	VBB supply undervoltage	Monitor
VRU	VREG output undervoltage	Monitor
VLRU	External supply regulator undervoltage	Monitor

The first bit (bit 15) of the Status register contains a common fault flag, FF, which will be high if any of the fault bits in the registers except IG have been set. In all cases the fault bits in the diagnostic registers are latched and only cleared when writing corresponding bit set 1 or states are reset. Some of the bit is a general flag associated with sub register so bottom flag need to be cleared.

Diagnostic registers

The diagnostic registers provide additional details of any VDS overvoltage, bootstrap undervoltage, VCP undervoltage or supply undervoltage faults. The status and diagnostic registers are described further in the serial interface section description below.

Chip-level diagnostics.

Chip-wide parameters critical for correct operation of the GDU are monitored. These include maximum chip temperature, minimum internal logic supply voltage, NVM data integrity and the serial interface transmission. These four monitors are necessary to ensure that the GDU is able to respond as specified.

Chip Fault State: Internal logic undervoltage (POR)

The GDU has an independent integrated logic regulator to supply the internal logic. This is to ensure that external events, other than loss of supply, do not prevent the GDU from operating correctly. The internal logic supply regulator will continue to operate with a low supply voltage, for example if the main supply voltage drops to a very low value during a severe cold crank event. In extreme low supply voltage circumstances, or during power-up or power-down, an undervoltage detector ensures that the GDU operates correctly. The logic supply undervoltage lockout cannot be masked as it is essential to guarantee correct operation over the full supply range.

When power is first applied to the GDU, the internal logic is prevented from operating, and all gate drive outputs held in the off state until the internal regulator voltage exceeds the logic supply undervoltage lockout threshold, derived from V_{BRR} . At this point all serial control registers will be reset to their power-on state and all fault states and the general fault flag will be reset. The FF bit and the POR bit in the Status register will be set to one to indicate that a power-on-reset has taken place. The GDU then goes into its fully operational state and begins operating as specified. In general the VBU, VRU and VLRU bits may also be set following a power-on-reset as the regulators may not have reached their respective rising undervoltage thresholds until after the register reset is completed.

When VLR rises above the rising undervoltage threshold, V_{LR_ris} , there are two possible sequences depending on the state of the WD mask bit.

If the WD mask bit is 0, i.e. watchdog is active, the MRSTn output will remain low for t_{WI} after the external logic supply,

VLR, rises above the rising undervoltage threshold, V_{LR_ris} . After t_{WI} MRSTn will go high and the watchdog timer will be reset. The watchdog timer will remain reset for t_{WI} during which time any valid Q&A watchdog transaction will be ignored. The gate drive outputs will remain disabled until the first valid watchdog is detected.

If the WD mask bit is 1, i.e. watchdog is inactive, the MRSTn output will remain low for t_{WI} , after the external logic supply, VLR, rises above the rising undervoltage threshold, V_{LR_ris} . The gate drive outputs can be activated t_{WI} after MRSTn goes high. A valid serial transfer will immediately reset this timeout, set MRSTn high and allow the gate drive outputs to be activated. The VLRU fault bit remains in the Diagnostic register until cleared.

Once the 89201 is operational, the internal logic supply continues to be monitored. If, during the operational state, VDD drops below logic supply undervoltage lockout threshold, derived from V_{BBR} , then the logical function of the GDU cannot be guaranteed and the outputs will be immediately disabled. The GDU will enter a power-down state and all internal activity, other than the logic regulator voltage monitor will be suspended. If the logic supply undervoltage is a transient event then the 89201 will follow the power-up sequence above as the voltage rises.

Chip Fault State: Over temperature

If the chip temperature rises above the over temperature threshold, T_{JF} , the general fault flag will be active and the over temperature bit, OT, will be set in the Status register. If FOT=1 in the Stop on Fail register when an over temperature is detected, all gate drive outputs will be disabled, the VLR regulator is shut down and MRSTn will go low. All GDU functions other than the VLR regulator remain active. When T_J drops by more than the overtemperature threshold ($T_J < T_{JF} - T_{JHyst}$) there are two possible sequences depending on the state of the WD mask bit.

If the WD mask bit is 0, i.e. watchdog is active, the VLR regulator will be re-enabled when T_J drops by more than the overtemperature threshold ($T_J < T_{JF} - T_{JHyst}$). MRSTn will remain low for t_{WI} after VLR rises above the rising undervoltage threshold, V_{LR_ris} . After t_{WI} MRSTn will go high and the watchdog timer will be reset. The watchdog timer will remain reset for t_{WI} during which time any valid Q&A watchdog transaction will be ignored. The gate drive outputs will remain disabled until the first valid watchdog is detected.

If the WD mask bit is 1, i.e. watchdog is inactive the VLR regulator will be re-enabled when T_J drops by more than the overtemperature threshold ($T_J < T_{JF} - T_{JHyst}$). MRSTn will remain low for t_{WI} after VLR rises above the rising undervoltage threshold, V_{LR_ris} . The gate drive outputs can be activated t_{WI} after MRSTn goes high. A valid serial transfer will immediately terminate this timeout, set MRSTn high and allow the gate drive outputs to be activated.

If FOT=0 in the Stop on Fail register when an over temperature is detected no circuitry will be disabled and action must be taken by the user to limit the power dissipation in some way so as to prevent over-temperature damage to the chip and unpredictable device operation.

In both cases the over temperature bit, OT, remains latched in the Status register until reset.

Chip Fault State: Serial Error

If any of the following errors are detected, then the transfer will be cancelled.

- Serial Time Out (ENV)
- Serial transaction while NVM is running (ESZ)
- Serial communication acknowledgement error (EAC)
- Serial communication CRC error (ECRC)

No data will be written to the registers. In addition, the Status register will not be reset, and the SE bit will be set to indicate a data transfer error. The diagnostics 1 register can be used to observe detailed flag associated.

Chip Fault State: NVM

Configuration and calibration information is stored within internal NVM and loaded into working registers to configure the device at power up. As part of this process a data integrity check is carried out. If the check returns a single bit error automatic error correction is applied and the part starts up. If the check returns a multiple bit error all gate drives are disabled, the general fault flag is set low and the NVM error bit, EE, is set in the Status register. NVM faults can only be cleared by a power on reset (POR).

Operational Monitors

Parameters related to the safe operation of the GDU in a system are monitored. These include parameters associated with external active and passive components, power supplies, and interaction with external controllers.

Voltages relating to driving the external power MOSFETs are monitored, specifically VREG, and each bootstrap capacitor voltage. The main supply voltage, VBB, is monitored for undervoltage events.

In addition, a watchdog timer can be applied to the ENABLE input to verify continued operation of the external controller. Q&A watchdog is also provided to monitor the MCU activity.

Monitor: VBB supply undervoltage

The main supply to the 89201 on the VBB terminal, VBB, is monitored to indicate if the supply voltage has dropped below its normal operating range (for example, during a load dump event).

If VBB falls below the VBB undervoltage warning threshold, V_{BBUV_fal} , then the VBU bit will be set in the status register. The general fault flag will be set and if FVBU=1 in the Stop on Fail register all gate drive outputs will be driven low.

When VBB moves above the rising VBB undervoltage threshold, V_{BBUV_ris} , the gate drive outputs will be reenabled. The VBU will remain set until it is reset.

If FVBU=0 in the Stop on Fail register the gate drive outputs will not be disabled.

Monitor: VREG undervoltage

The internal charge-pump regulator supplies the low side gate driver and the bootstrap charge current. It is critical to ensure that the regulated voltage, V_{REG} , at the VREG terminal is sufficiently high before enabling any of the outputs.

If V_{REG} goes below the VREG undervoltage threshold, V_{ROFF} , the general fault flag will be active, the VREG under voltage bit, VRU, will be set in the Status register and all gate drive outputs will go low. When VREG rises above the rising threshold, V_{RON} , the gate drive outputs are re-enabled. The fault bit remains in the Status register until cleared.

The VREG undervoltage monitor circuit is active during power up. The general fault flag will be active, and all gate drives will be low until VREG is greater than V_{RON} . Note that this is sufficient to turn on standard threshold external power MOSFETs at a battery voltage as low as 5.5V but the on-resistance of the MOSFET may be higher than its specified maximum.

The VREG undervoltage monitor can be disabled by setting the VRU bit in the Mask register. Although not recommended, this can allow the GDU to operate below its minimum specified supply voltage level with a severely impaired gate drive. The specified electrical parameters will not be valid in this condition.

Monitor: VCP Under voltage

The internal charge-pump regulator generates VCP voltage, which is above VBRG. It is critical to ensure the VBRG voltage to drive high-side MOSFET due to bootstrap management.

If VCP goes below VCP under voltage threshold, V_{CPUV} , the general fault flag will be active, the VCP undervoltage bit, VCPU, will be set in the Status register and all gate drive output will go low. When VCP rises above the rising threshold, $V_{CPU} + V_{CPU_{hys}}$, the gate drive outputs are reenabled. The fault bit remains in the Status register until cleared.

Monitor: VLR Under voltage

The voltage at the VLR terminal, V_{LR} , is monitored to ensure that the supply for any external controller is high enough to permit correct operation of the controller. If VLR drops below the falling undervoltage threshold, V_{LRUV_fal} , the regulator undervoltage bit, VLRU, will be set in the Status register and the MRSTn output will go low to reset the external microcontroller. If FVLRU = 1 in the Stop on Fail register, all gate drive outputs will be disabled. It will set the VLRU bit in the Status register.

When VLR rises above the rising undervoltage threshold, V_{LRUV_ris} , there are two possible sequences depending on the state of the WD mask bit.

If the WD mask bit is 0, i.e. watchdog is active, the MRSTn output will remain low for t_{WI} after the external logic supply, VLR, rises above the rising undervoltage threshold, V_{LRUV_ris} .

After t_{WI} , MRSTn will go high and the watchdog timer will be reset. The watchdog timer will remain reset for t_{WI} during which time any Q&A watchdog transaction will be ignored. The gate drive outputs will remain disabled until the first valid watchdog is detected.

If the WD mask bit is 1, i.e. watchdog is inactive, the MRSTn output will remain low for t_{WI} , after the external logic supply, VLR, rises above the rising undervoltage threshold, V_{LRUV_ris} . The gate drive outputs can be activated t_{WI} after MRSTn goes high. A valid serial transfer will immediately terminate this timeout, set MRSTn high and allow the gate drive outputs to be activated. The VLRU fault bit remains in the Status register until cleared.

Monitor: Temperature warning

If the chip temperature rises above the temperature warning threshold, T_{JWH} , the general fault flag will be active and the hot warning bit, TW, will be set in the Status register. No action will be taken by the GDU. When the temperature drops below T_{JWH} by more than the hysteresis value, T_{JWHys} , the general fault flag is reset but the TW bit remains in the Status register until reset.

Monitor: Q&A watchdog

The programmable watchdog can be used to determine if an external microcontroller is operating in an adverse state. When a watchdog fault is detected, the fault state is latched, the WD bit is set in the Status register and the MRSTn output goes active low for t_{WI} in order to reset the microcontroller. When MRSTn goes high after the t_{WI} low period the watchdog timer will be reset and remain reset for t_{WI} . The fault state is cleared and the GDU will re-enable the gate drive outputs on the first valid watchdog transition after the t_{WI} timeout. The WD fault bit remains set until cleared. Further details of the microcontroller watchdog operation are provided in the Q&A Watchdog section.

Monitor: ENABLE timeout

The ENABLE input provides a direct connection to all gate drive outputs and can be used as a safety override to immediately disable the outputs. The ENABLE input can be programmed to operate as a direct logic control by setting the ETO mask bit to 1 or can be monitored by a watchdog timer by setting ETO to 0. In the direct mode the input is not monitored. In the timed mode, the first change of state on the ENABLE input will activate the gate drive outputs under command from the corresponding phase control signals and a timer is started. The ENABLE input must then change state before the end of the ENABLE timeout period, t_{ETO} . If the ENABLE input does not change before the end of the timeout period, then all gate drive outputs will be driven low, the ETO bit will be set in the Status register and the general fault flag will be active. Any following change of state on the ENABLE input will re-activate the gate drive outputs and reset the general fault flag. The ETO bit remains in the Status register until reset.

Monitor: IG input

The state of the IG input is provided directly in the status register as the IG bit. This bit has no effect on the state of the FF bit.

Power Bridge Faults**Bridge: Bootstrap capacitor undervoltage fault**

The GDU monitors the individual bootstrap capacitor charge voltages to ensure sufficient high-side drive.

The action taken when a valid bootstrap undervoltage fault is detected and the fault reset conditions depend on the state of the FBU bit in the Stop on Fail register.

If FBU= 0 in the Stop on Fail register, the fault state will be latched, the general fault flag and BU will be active in the status register, the associated bootstrap undervoltage fault bit (VA, VB, VC) will be set in the Diagnostic 1 register and the associated MOSFET will be disabled. The fault state, but not the bootstrap undervoltage fault bit, will be reset the next time the MOSFET is commanded to switch on. If the MOSFET is being driven with a PWM signal then this will usually mean that the MOSFET will be turned on again each PWM cycle. If this is the case, and the fault conditions remains, then a valid fault will again be detected after the timeout period and the sequence will repeat. In this case the general fault flag will only be reset for the duration of the validation timer. The bootstrap undervoltage fault bit will only be reset by a serial read of the Diagnostic register or by a power-on reset.

If FBU= 1 in the Stop on Fail register, the fault will be latched, the general fault flag and BU will be active, the associated bootstrap undervoltage fault bit (VA, VB, VC) will be set in the Diagnostic register 1 and all MOSFETs will be disabled. The fault state, bootstrap undervoltage fault bit and the general fault flag will only be reset by a serial read of the Diagnostic register or by a power-on reset.

The bootstrap undervoltage monitor can be disabled for all phases by setting the BSU bit in the Mask register. Although not recommended, this can allow the GDU to operate below its minimum specified supply voltage level with a severely impaired gate drive. The specified electrical parameters may not be valid in this condition.

Bridge: MOSFET VDS overvoltage fault

Faults on any external MOSFETs are determined by monitoring the drain-source voltage of the MOSFET and comparing it to a drain-source overvoltage threshold. The threshold, V_{DST} , is generated by an internal DAC and is defined by the values in the VT[5:0] variable. This variable provides the input to a 6-bit DAC with a least significant bit value of typically 50mV. The output of the DAC produces the threshold voltage approximately defined as:

$$V_{DST} = n \times 50mV$$

where n is a positive integer defined by VT[5:0]

The low-side drain-source voltage for any MOSFET is measured between the adjacent Sx terminal and the adjacent LSS terminal. Using the LSS terminal rather than the ground connection avoids adding any low-side current sense voltage to the real low-side drain-source voltage and avoids false VDS fault detection.

The high-side drain-source voltage for any MOSFET is measured between the adjacent Sx terminal and the VBRG terminal. Using the VBRG terminal rather than the VBB avoids adding any reverse diode voltage or high-side current sense voltage to the real high-side drain-source voltage and avoids false VDS fault detection.

The VBRG terminal is an independent low-current sense input to the top of the MOSFET bridge. It should be connected independently and directly to the common connection point for the drains of the power bridge MOSFETs at the positive supply connection point in the bridge. The input current to the VBRG terminal is proportional to the drain-source threshold voltage, V_{DSTH} , and is approximately:

$$IVBRG = 0.84VBRG + 20VDST + 107$$

where $IVBRG$ is the current into the VBRG terminal in μA and $VDST$ is the drain-source threshold voltage described above.

The output from each VDS overvoltage comparator is filtered by a VDS fault qualifier circuit. This circuit uses a timer to verify that the output from the comparator is indicating a valid VDS fault. The duration of the VDS fault qualifying timer, t_{VDQ} , is determined by the contents of the VQT[5:0] variable. t_{VDQ} is approximately defined as:

$$t_{VDQ} = n \times 100ns$$

where n is a positive integer defined by VQT[5:0] The qualifier can operate in one of two ways, debounce mode, or blanking mode, selected by the VDQ bit.

In debounce mode (the default setting) a timer is started each time the comparator output indicates a VDS fault detection when the corresponding MOSFET is active. This timer is reset when the comparator changes back to indicate normal operation. If the debounce timer reaches the end of the timeout period, set by t_{VDQ} , then the VDS fault is considered valid and the corresponding VDS fault bit, ALO, AHO, BLO, BHO, CLO or CHO will be set in the Diagnostic register 1.

In blanking mode (optional) a timer is started when any gate drive is turned on or turned off. The outputs from the VDS overvoltage comparators for all MOSFETs are ignored (blanked) for the duration of the timer's active period, set by t_{VDQ} . If any gate drive changes state while a blanking period is in progress the timer will be re-triggered, resulting in an extended overall blanking time. If any comparator output indicates a VDS fault and the blanking timer is not active then the VDS fault is considered valid and the corresponding VDS fault bit, ALO, AHO, BLO, BHO, CLO or CHO is set in the Diagnostic register 1.

The action taken when a valid VDS fault is detected, and the action then required to clear the fault state depend upon the FDSO bit in the Stop on Fail register value.

If FDSO= 0 in the Stop on Fail register, the fault state will be latched, the associated VDS fault bit will be set and the associated phase will be disabled. The fault state, but not the VDS fault bit, will be reset the next time the MOSFET on which the fault was detected is commanded to switch on. If the phase is being driven with a PWM signal then this will usually mean that the phase will be turned on again each PWM cycle. If this is the case, and the fault condition remains, then a valid fault will again be detected after the timeout period and the

sequence will repeat. In this case the general fault flag will only be reset for the duration of the validation timer. The VDS fault bit will only be reset by writing 1 to corresponding bit or by a power-on reset.

If FDSO= 1 in the Stop on Fail register, the fault will be latched, the associated VDS fault bit will be set and all MOSFETs will be disabled. The fault state, VDS fault bit and the general fault flag will only be reset by a serial read of the Diagnostic register with writing 1 to corresponding bit or by a power-on reset.

If FDSO= 0 in the Stop on Fail register, care must be taken to avoid damage to the MOSFET where the VDS fault is detected. Although the MOSFET will be switched off as soon as the fault is detected at the end of the fault validation timeout, it is possible that it could still be damaged by excessive power dissipation and heating. To limit any damage to the external MOSFETs or the motor, the MOSFET should be fully disabled by logic inputs from the external controller.

Fault action

The action taken when one of the diagnostic functions indicates a fault is listed in table 8.

When a fault is detected a corresponding fault state is considered to exist. In some cases the fault state only exists during the time the fault is detected. In other cases, when the fault is only detected for a short time, the fault state is latched (stored) until reset. The faults that are latched are indicated in table 8. Some latched fault states are reset with specific actions. All latched fault states and any fault bits in the status diagnostic registers are always reset when a poweron-reset state is present or when the associated fault bit is cleared.

For many of the faults the action taken is modified by the SoF (Stop on Fault) register. In general this will affect the action on the gate drive outputs and is detailed for each fault in table 8. Setting the corresponding SoF register bit to 0 such that the gate drive outputs are not disabled in the event of the corresponding fault being detected means that the GDU will not take any action to protect itself or the external bridge MOSFETs and damage may occur.

Appropriate action must be taken by the external controller.

The fault conditions power-on-reset, VREG undervoltage and NVM error are considered critical to the safe operation of the GDU and the system. If these faults are detected, then the gate drive outputs are automatically driven low and

Table 8: Fault Actions

Fault Description	SoF Bit	Disable Output		Fault State Latched
		if 0	if 1	
No Fault	-	No	No	-
Power on-reset	-	Yes ¹	Yes ¹	No
VREG Undervoltage	-	Yes ¹	Yes ¹	No
VBB Undervoltage	FVBU	No	Yes ¹	No
VLR Undervoltage	FVLRU	No	Yes ¹	Yes
VCP Undervoltage	FVCPU	No	Yes ¹	No
Q&A Watchdog	FWD	No	Yes ¹	Yes
ENABLE Timeout	FEN	No	Yes ¹	No
VDS Overvoltage	FDSO	Yes ²	Yes ¹	Yes
Bootstrap Undervoltage	FBU	Yes ²	Yes ¹	Yes
Over Temperature	FOT	No	Yes ¹	Yes ³
Temperature Warning	FTW	No	Yes ¹	No
Serial Transmission Error	FSE	No	No	No
NVM	-	Yes ¹	Yes ¹	Yes

1: All gate drives low, all MOSFETs off

2: Gate drive to the affected MOSFET low, only the affected MOSFET off

3: all MOSFETs in the bridge held in the off state. This state will remain until the fault is removed.

If the ENABLE timeout is active by setting the ETO mask bit to 0 then this fault state is also considered critical to the safe operation of the GDU and the system. If an ENABLE timeout is detected then all gate drive outputs

are driven low and all MOSFETs in the bridge held in the off state. This state will remain until the timer is reset.

If the microcontroller watchdog is active by setting the WD mask bit to 0 then this fault state is also considered critical to the safe operation of the GDU and the system. If a microcontroller watchdog failure is detected then all gate drive outputs are driven low and all MOSFETs in the bridge held in the off state. This state will remain until a valid Q&A watchdog transaction has been detected the fault state is cleared.

For the overtemperature fault conditions the action taken depends on the status of the FOT on SoF. If an overtemperature fault is detected and FOT=1 on Stop on Fail register when an over temperature is detected, all gate drive outputs will be disabled, the VLR regulator is shut down and MRSTn will go low. This state will remain until t_{WI} after the fault is removed. If FOT =1 on SoF then neither VLR nor the gate drive outputs will be affected.

If corresponding FOT=0 on SoF when an over temperature is detected no circuitry will be disabled and action must be taken by the user to limit the power dissipation in some way so as to prevent over-temperature damage to the chip and unpredictable device operation.

VBB undervoltage is not considered critical and the action taken depends on the state of FVBU on the Stop on Fail register. If a VBB undervoltage condition exists and FVBU =1 on SoF the gate drive outputs will be disabled, and the motor will coast until the condition is removed. IF FVBU=0 on SoF the the gate drive outputs are not affected.

If a VDS overvoltage or bootstrap undervoltage fault is detected then the action taken will also depend on the status of the corresponding bit on SoF but these faults are handled as a special case. If a fault is detected on any of these two diagnostics and corresponding bit is 1 on SoF then all the gate drive outputs will be driven low and all MOSFETs in the bridge held in the off state. When corresponding bit is 1 on SoF this fault state will be latched and remain until reset. If a VDS overvoltage or bootstrap undervoltage fault is detected and corresponding bit is 0 on SoF then only the gate drive output to the MOSFET where the fault was detected will be driven low and the MOSFET held in the off state. When corresponding bit is 0 on SoF the VDS or bootstrap undervoltage fault state will be latched but will be reset the next time the MOSFET is commanded to switch on.

The remaining faults; temperature warning, overcurrent detect and serial error, are not considered critical and the gate drive outputs will remain active.

Fault Masks

Individual diagnostics, except power-on reset, serial transmission error and NVM error can be disabled by setting the corresponding bit in the mask registers. Power-on-reset cannot be disabled because the diagnostics and the output control depend on the logic regulator to operate correctly. If a bit is set to one in the mask registers then the corresponding diagnostic will be completely disabled. No fault states for the disabled diagnostic will be generated, no action will be taken and no fault flags or diagnostic bits will be set. See Mask Register definition for bit allocation.

Care must be taken when diagnostics are disabled to avoid potentially damaging conditions.

Diagnostic and System Verification

To comply with various aspects of safe system design it is necessary for higher-level safety systems to verify that any diagnostics or functions used to guarantee safe operation are operating within specified tolerances.

The GDU includes additional verification functions to help the system design comply with safety requirements.

Table 9: Verification Functions

Verification Type	Function Verified	Operation	
		Off-line	On-line
Monitor	System clock frequency verification		Y
Diagnostic	System clock operation		Y
Diagnostic	All gate drives off	Y	
Monitor	MOSFET active monitor	Y	

Online Verification

The following functions are permanently active and can be monitored at any point during normal operation. No other action will be taken by the GDU unless described below.

System Clock Operation Verification

The internal system clock is monitored by a watchdog timer. When the watchdog detects that the clock has stopped, the gate drive outputs are disabled. If a serial access is performed, data input through SDI is ignored, no data is written to the serial registers and the addresses registers are clocked out on the SDO terminal. If the internal system clock stops then the SC bit will be set in the Diagnostics register 0. When the system clock restarts the SC bit will be reset and the gate drive outputs and the serial communication reenabled.

Offline Verification

The following functions can only be completed when the diagnostics are not required and the motor is not being commanded to run.

All Gate Drives Off

The successful propagation of control inputs demanding all gate drives-off to the gate drive outputs is verified by setting up an appropriate input condition and inspecting the GDO bit in the Diagnostic 0 register. If the input condition has successfully turned off all six gate drives, the GDO bit is set. The control input conditions (i.e. the combinational states of Hx, Lx, xH, xL, and ENABLE) that demand all outputs off ($GHx = L$, $GLx = L$) and hence set the GDO bit as a result of a successful verification test can be determined by inspection of tables 2, 3, and 4. If the ENABLE watchdog mode is selected ($EWD = 1$) and the watchdog timeout is allowed to expire, the GDO bit will similarly be set as the result of a successful test. Verification of propagation from an appropriate combination of phase logic inputs (Hx, Lx) and serial register bits (xH, xL) to the gate drive outputs does not verify propagation from the ENABLE input to the gate drive outputs and vice versa. Gate drive off events are not latched in the Diagnostics register and the GDO bit returns to 0 as soon as any gate drive is detected to be in the on state.

MOSFET Active Monitor

The state of the external MOSFETs in each phase can be verified using a combination of MOSFET commands.

For the high-side MOSFET test, a high-side MOSFET is switched on using the Control register bits or the logic input terminals. The MOSFET active monitor then determines whether the resultant voltage on the corresponding Sx node is higher than $V_{BRG} - V_{DSTH}$. If this is true, the corresponding bit in the Diagnostics 0 register will be set to 1. If the bit is set to 0 then the high-side MOSFET verification has failed.

For the low-side MOSFET test, a low-side MOSFET is switched on using the Control register bits or the logic input terminals. The MOSFET active monitor then determines whether the resultant voltage on the corresponding Sx node is lower than $L_{SS} + V_{DSTL}$. If this is true the corresponding bit in the Diagnostics 0 register will be set to 0. If the SxM bit is set to 1 then the low-side MOSFET verification has failed.

All six MOSFETs must be tested separately to complete the verification.

Non-volatile memory

The values in the configuration and control registers are held in non-volatile NVM allowing the GDU to be preprogrammed with different user defined register values for each application thus avoiding the need to program the register contents at each power on.

The GDU provides a simple method to write the contents of the registers into the NVM using the serial interface. When the WNV[1:0] bits in register 121 are changed from [01] to [10], in a single serial write, the present contents of registers from the address 0 to 40 and 123 are saved (written to NVM) as a single operation. The save sequence takes typically 400ms to complete. It is not possible to save single register values. Although the motor may be operating during the save sequence, it is recommended that the motor drive is disabled before starting a save sequence to avoid any corruption caused by the electrical noise or any faults from the motor.

The necessary voltage to writing NVM is internally generated. When the save command is accepted, GDU automatically generate a programming voltage and complete the sequence.

During the save sequence the I2C remains active for read only. Any attempt to write to the registers during the save sequence will cause the ESZ in the diagnostics 1 register and corresponding register FF and SE bits to be set in the Status register. ESZ need to be cleared to reset the flag in the status register.

During the save sequence the GDU will automatically complete all the necessary steps to ensure that the NVM is correctly programmed and will complete the sequence by verifying that the contents of the NVM have been securely programmed. On successful completion of a save sequence the WNV[1:0] bits will be set to 01. Register 24 should be read to determine if the save has completed successfully. If WNV[1:0] is reset to 00 then the save sequence has been terminated and has not completed successfully.

To externally verify the data saved in the NVM the VBB supply must be cycled off then on to cause a power-on reset. Following a power-on reset the contents of the NVM are copied to the serial registers when can then be read through the serial interface and verified.

For guaranteed data retention reliability, the NVM in the GDU should be written no more than 1000 times.

06/Jan/2022	Initial Revision
19/Jan/2022	- Updated Frequency Table for I2C Connection - Updated DAU connection for VBIAS in voltage mode
24/Mar/2023	- Changed the typo in 'Regisiter' text to 'Register' - Corrected the typo in showing the start address for 'User Data area Register' as '0x09' instead of '0x08'
31/July/2024	- Fixed the VDS monitor typo in the block diagram in functional description section - Corrected the typo in product ID from A892011 to A89201
25/Nov/2024	- Updated default values for registers - Updated TCF and Rx register to read only
20/Feb/2026	- Changed term 'SDG' to 'DAG'

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